



Grayson White

1

Announcements

- Office Hours Schedule
- P-Set 1 available now.
 - Will discuss how to access the p-sets today!
 - Due at 9am the following Thursday
- My Undergraduate Forestry Data Science summer research program application deadline is **THIS FRIDAY!**

Week 2 Goals

Mon Lecture

- Basics of `ggplot2`
- Explore several `geoms`.
- And a little data wrangling with `dplyr` as needed!

Wed Lecture

- GitHub workflow overview
- Learn how to ask coding questions well.
- Graphing context!
 - Labels
 - Highlighting
 - Useful text
- Look at more `geoms`.
- Explore further customizations.
 - Color
 - Themes



P-set tips

- Start on your problem sets **early**, and **ask questions** (in Slack, office hours, the internet) when you get stuck!
- Problem sets are meant to be engaged with over multiple days. It is a lot less effective for your learning if you try to sprint through it the day before it is due.
- If/when you get stuck, it can be helpful to take a break! Some things to try:
 - Go onto another problem, solving it may give you ideas as to how to solve the one you are stuck on.
 - Think about the problem as you are brushing your teeth or laying in bed.
 - Time where you are “bored”/giving your brain time to wander can be helpful for problem solving.



Now: GitHub workflow demo

Recap Data: Births2015

```
1 # Load libraries
2 library(mosaicData)
3 library(tidyverse)
4
5
6 # Grab data
7 data(Births2015)
8
9 # Inspect data
10 glimpse(Births2015)
```

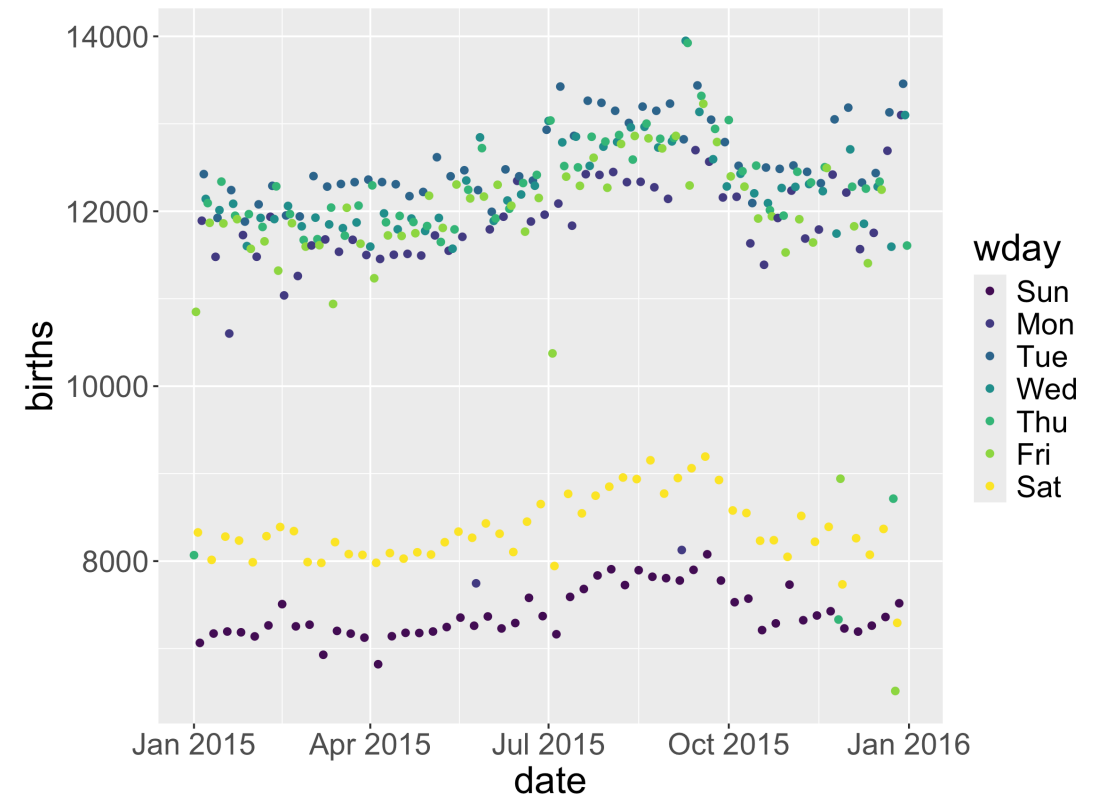
Rows: 365

Columns: 8

```
$ date      <date> 2015-01-01, 2015-01-02, 2015-01-03, 2015-01-04, 2015-01-...
$ births    <dbl> 8068, 10850, 8328, 7065, 11892, 12425, 12141, 12094, 1186...
$ wday      <ord> Thu, Fri, Sat, Sun, Mon, Tue, Wed, Thu, Fri, Sat, Sun, Mo...
$ year      <dbl> 2015, 2015, 2015, 2015, 2015, 2015, 2015, 2015, 2015, 201...
$ month     <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ...
$ day_of_year <int> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17...
$ day_of_month <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17...
$ day_of_week <dbl> 5, 6, 7, 1, 2, 3, 4, 5, 6, 7, 1, 2, 3, 4, 5, 6, 7, 1, 2, ...
```

Recap Data: Births2015

```
1 ggplot(data = Births2015,  
2       mapping = aes(x = date, y = births,  
3                     color = wday)) +  
4   geom_point()
```

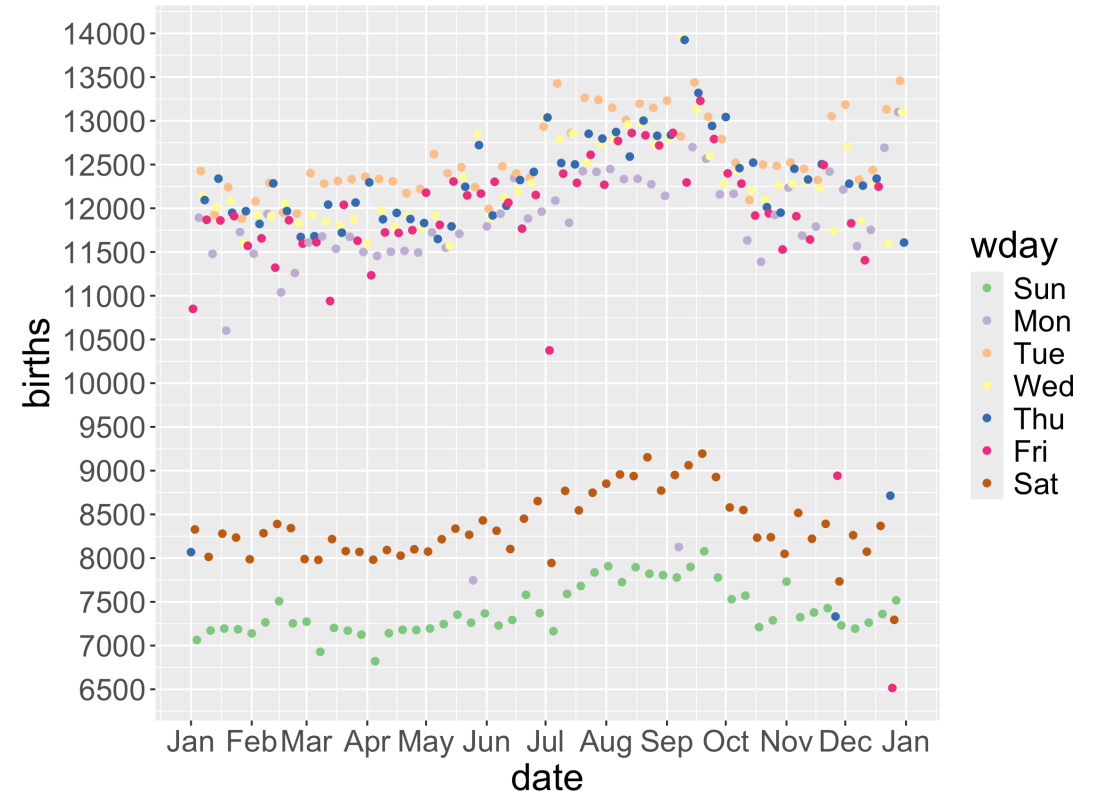


- Let's think more about the **scales**.
- Let's add more **context**!



Scales

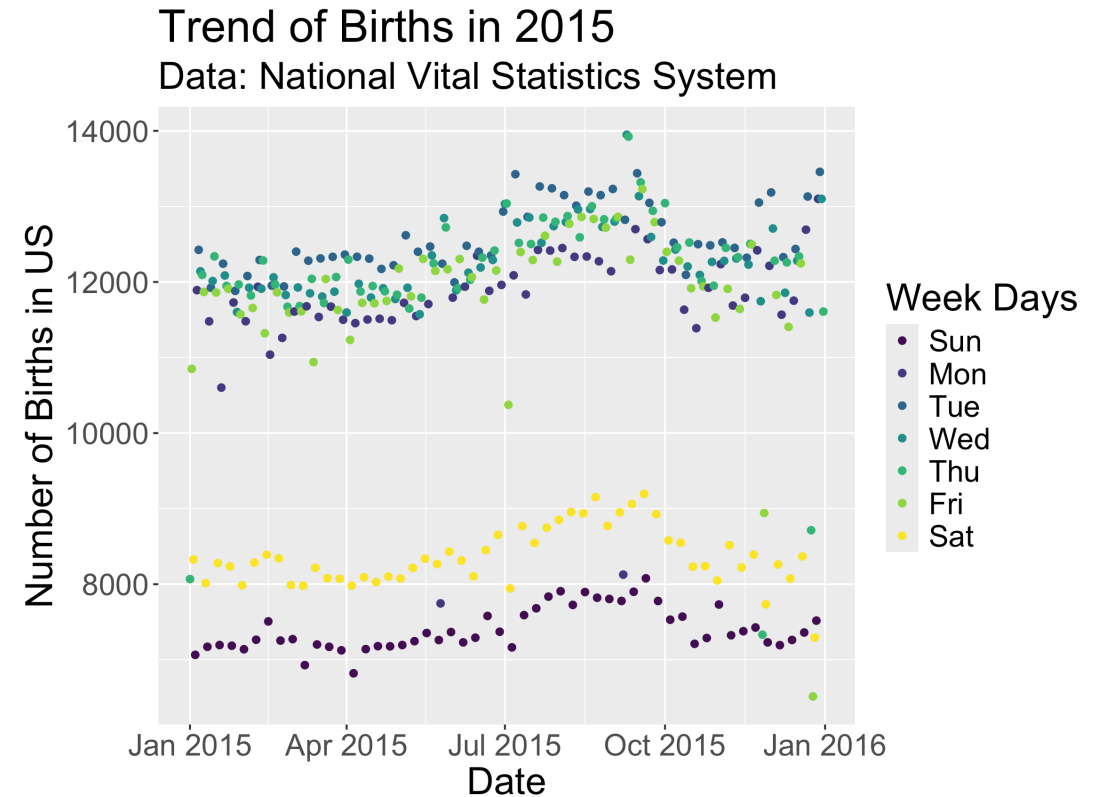
```
1 ggplot(data = Births2015,  
2       mapping = aes(x = date, y = births,  
3                     color = wday)) +  
4   geom_point() +  
5   scale_x_date(date_labels = "%b",  
6               date_breaks = "1 month") +  
7   scale_y_continuous(breaks = seq(6000,  
8                             14000, by = 500)) +  
9   scale_color_brewer(type = "qual",  
10                    palette = 1)
```



- Maybe we want to change the default settings of a scale.
- Maybe we want a different scale than the default.

Context: Labels

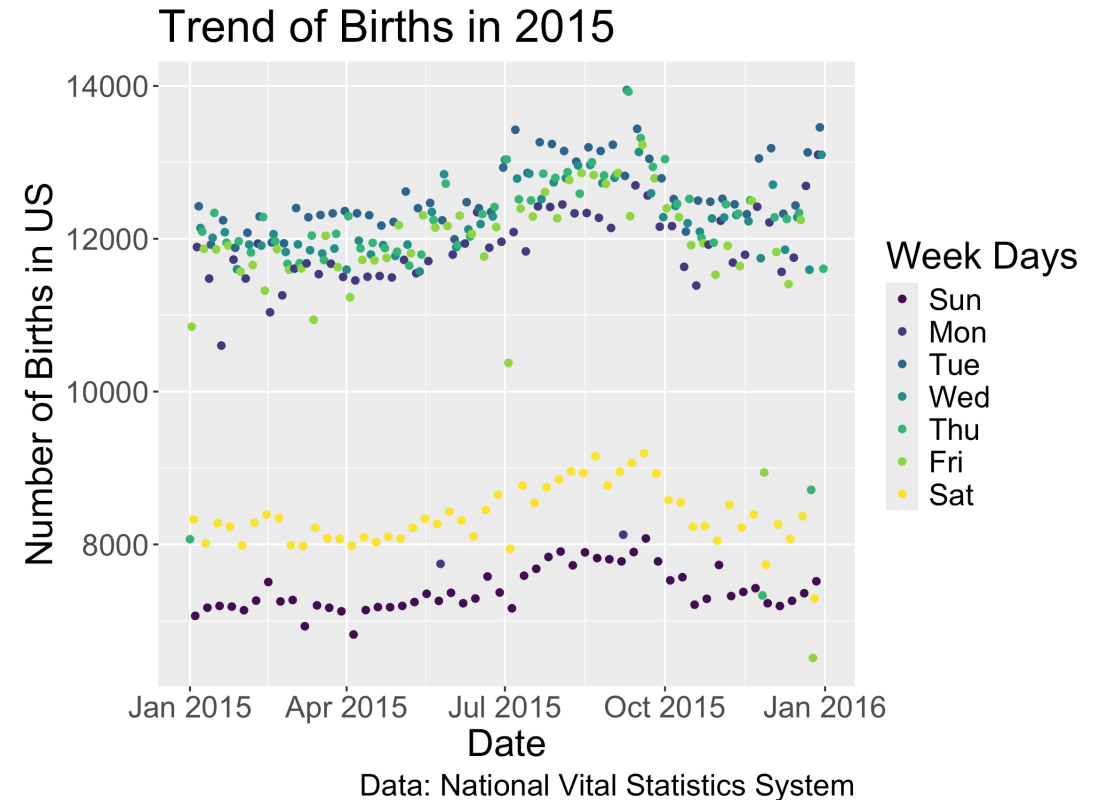
```
1 ggplot(data = Births2015,  
2       mapping = aes(x = date, y = births,  
3                     color = wday)) +  
4   geom_point() +  
5   labs(  
6     x = "Date",  
7     y = "Number of Births in US",  
8     title = "Trend of Births in 2015",  
9     subtitle = "Data: National Vital Statistics System",  
10    color = "Week Days"  
11  )
```



- Prefer citing the data at the bottom?

Context: Labels

```
1 ggplot(data = Births2015,  
2       mapping = aes(x = date, y = births,  
3                     color = wday)) +  
4   geom_point() +  
5   labs(  
6     x = "Date",  
7     y = "Number of Births in US",  
8     title = "Trend of Births in 2015",  
9     caption = "Data: National Vital Statistics System",  
10    color = "Week Days"  
11  )
```



- Prefer citing the data at the bottom?
- For slide space, I will neglect my labeling in the for the rest of the slides.
- Now we want to add even more context to help the reader understand whether or not birth rates on national holidays behave like weekends.



Context: Adding Holidays

```
1 library(lubridate)
2 holidays <-
3   data.frame(date = ymd("2015-01-01", "2015-05-25", "2015-07-04",
4     "2015-12-25", "2015-11-26", "2015-12-24",
5     "2015-09-07"),
6     occasion = c("New Year", "Memorial Day",
7       "Independence Day", "Christmas",
8       "Thanksgiving", "Christmas Eve",
9       "Labor Day"),
10    emoji = c("1f389", "1f396", "1f386", "1f384",
11      "1f983", "1f381", "1f477"))
12
13 holidays <- left_join(holidays, Births2015)
```

Context: Adding Holidays

```
1 glimpse(holidays)
```

Rows: 7

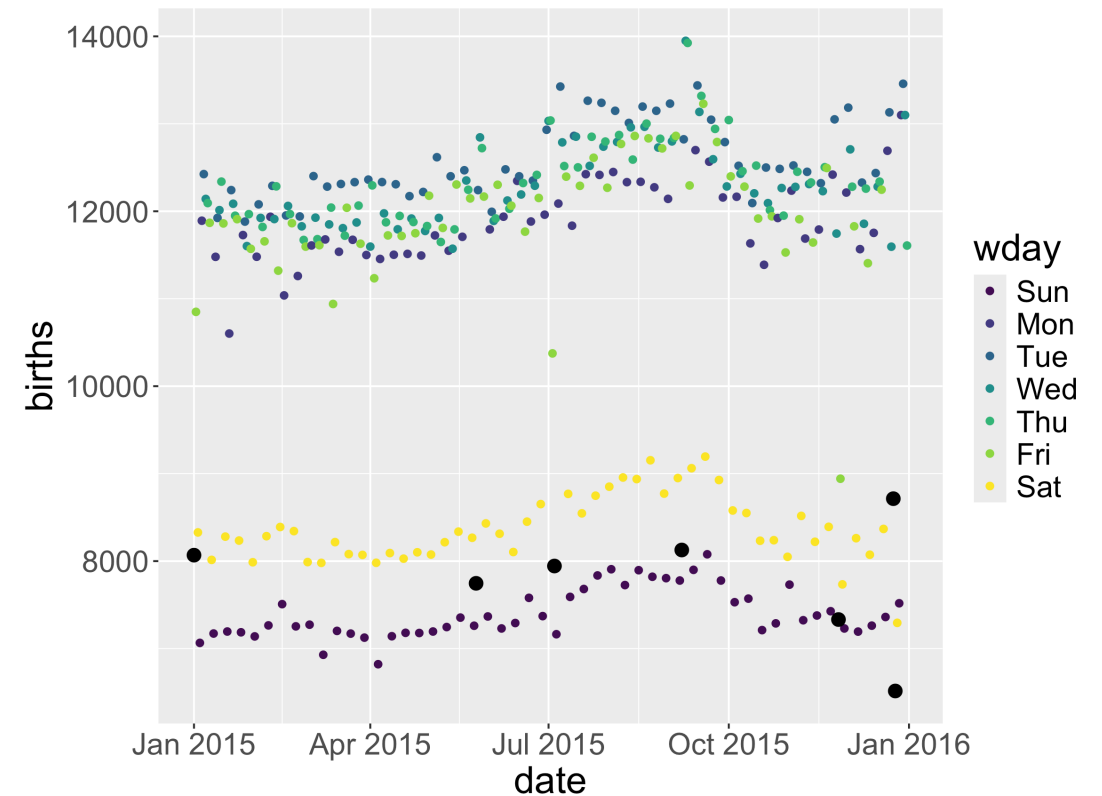
Columns: 10

```
$ date      <date> 2015-01-01, 2015-05-25, 2015-07-04, 2015-12-25, 2015-11-...
$ occasion  <chr> "New Year", "Memorial Day", "Independence Day", "Christma...
$ emoji     <chr> "1f389", "1f396", "1f386", "1f384", "1f983", "1f381", "1f...
$ births    <dbl> 8068, 7746, 7944, 6515, 7332, 8714, 8127
$ wday      <ord> Thu, Mon, Sat, Fri, Thu, Thu, Mon
$ year      <dbl> 2015, 2015, 2015, 2015, 2015, 2015, 2015
$ month     <dbl> 1, 5, 7, 12, 11, 12, 9
$ day_of_year <int> 1, 145, 185, 359, 330, 358, 250
$ day_of_month <dbl> 1, 25, 4, 25, 26, 24, 7
$ day_of_week <dbl> 5, 2, 7, 6, 5, 5, 2
```

- Let's add some context.

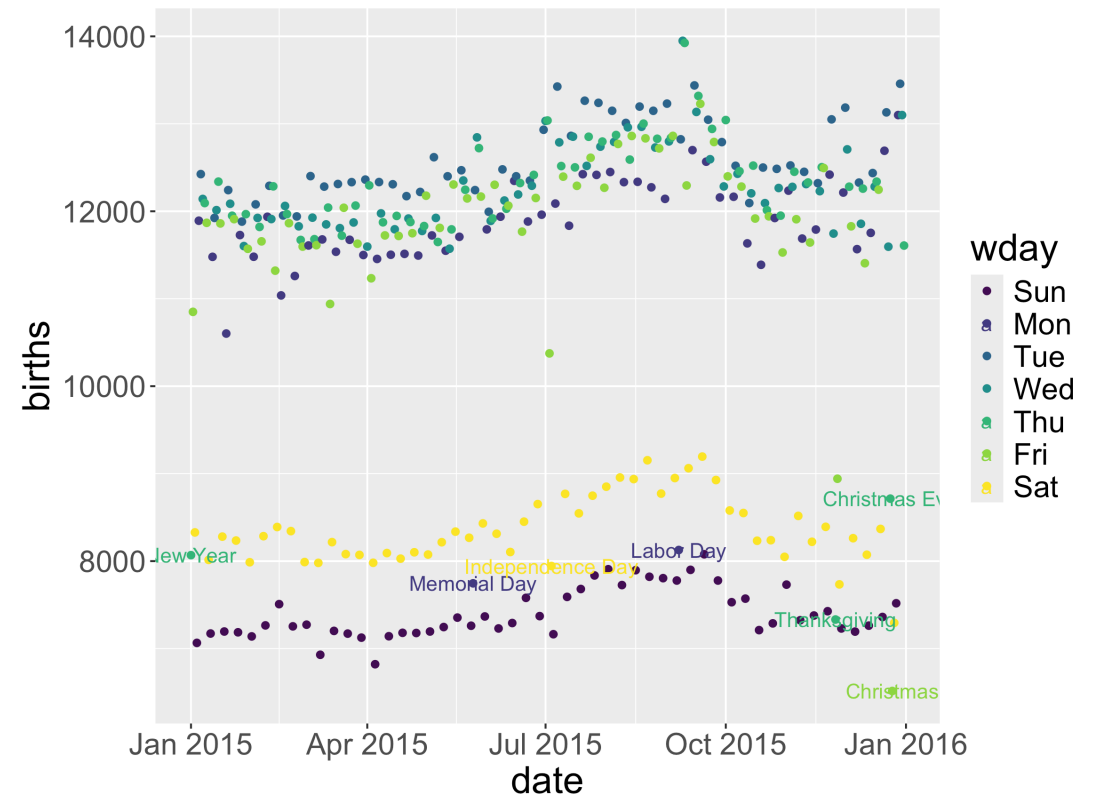
Context: Adding Holidays

```
1 ggplot(data = Births2015,  
2       mapping = aes(x = date, y = births,  
3                     color = wday)) +  
4   geom_point() +  
5   geom_point(data = holidays,  
6             mapping = aes(x = date, y = births),  
7             color = "black", size = 3)
```



Context: Adding Holidays

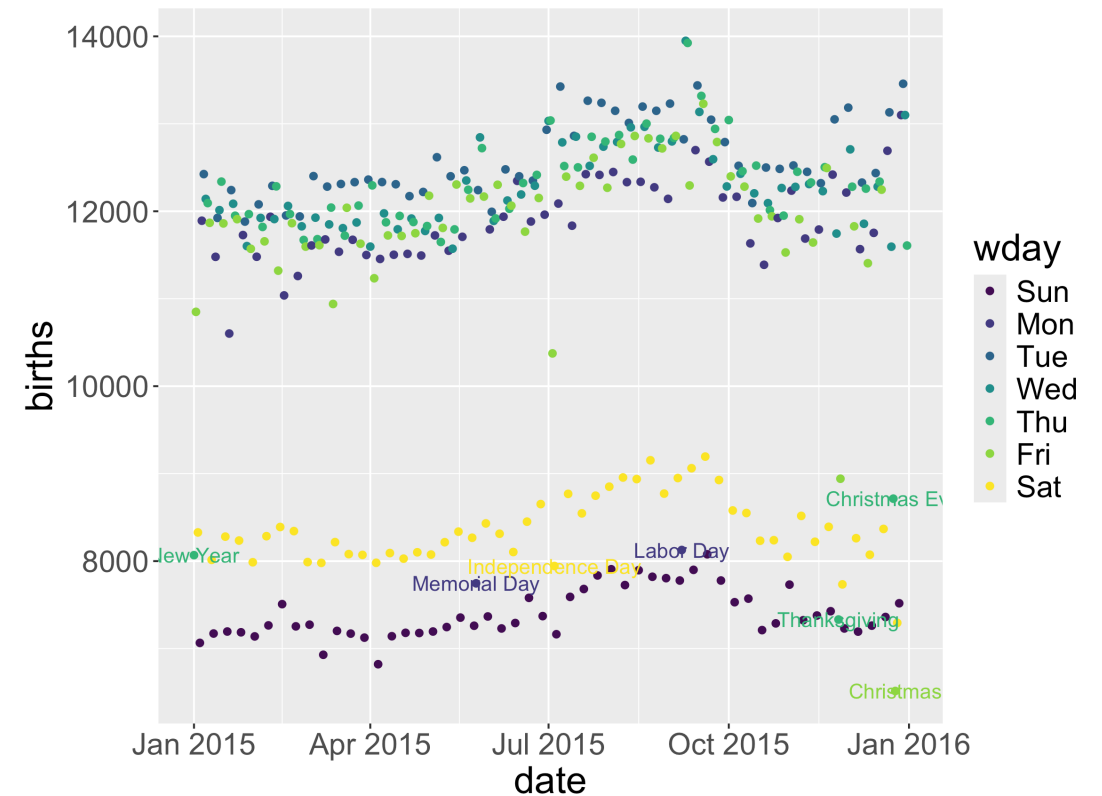
```
1 ggplot(data = Births2015,  
2       mapping = aes(x = date, y = births,  
3                     color = wday)) +  
4   geom_point() +  
5   geom_text(data = holidays,  
6             mapping = aes(label = occasion))
```



- Problems?

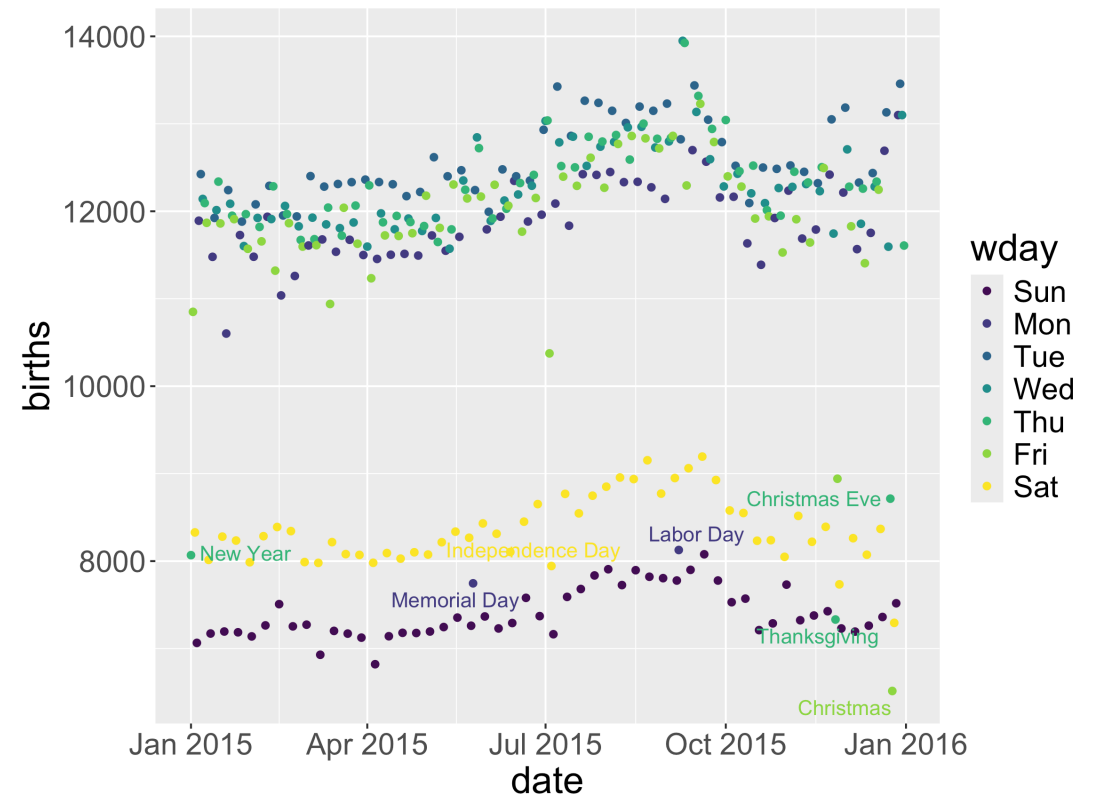
Context: Adding Holidays

```
1 ggplot(data = Births2015,  
2       mapping = aes(x = date, y = births,  
3                     color = wday)) +  
4   geom_point() +  
5   geom_text(data = holidays,  
6             mapping = aes(label = occasion),  
7             show.legend = FALSE)
```



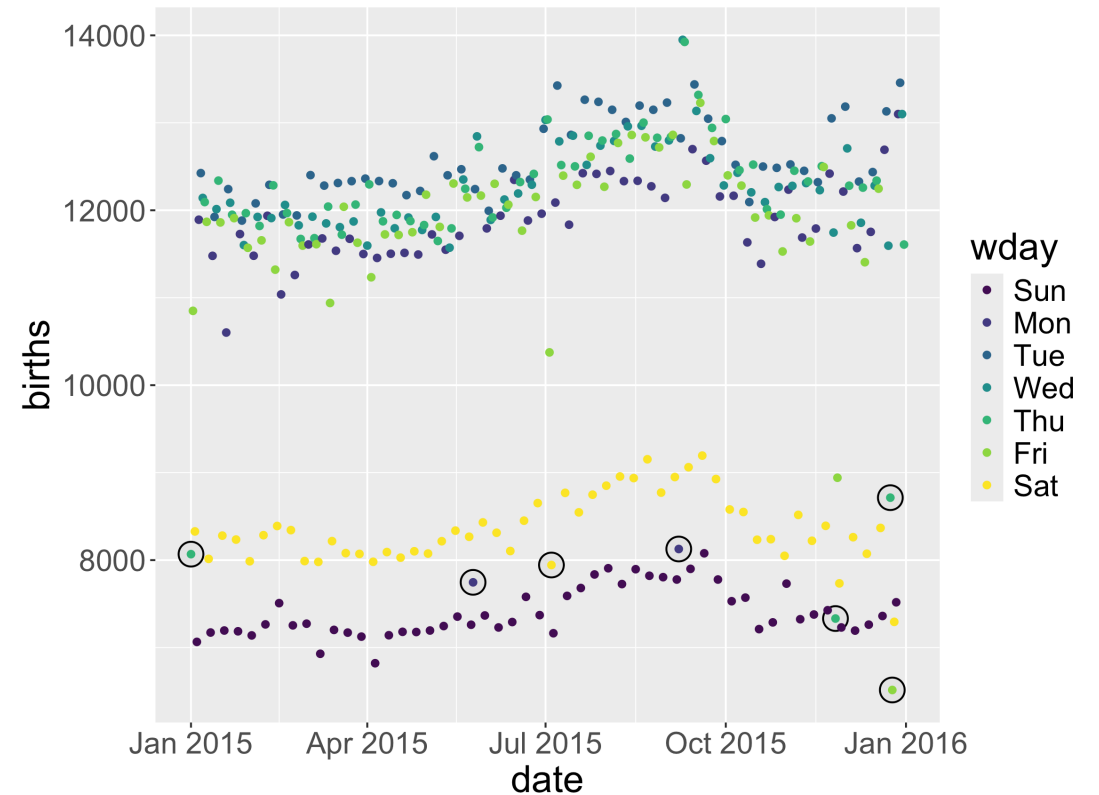
Context: Adding Holidays

```
1 library(ggrepel)
2 ggplot(data = Births2015,
3       mapping = aes(x = date, y = births,
4                     color = wday)) +
5   geom_point() +
6   geom_text_repel(data = holidays,
7                   mapping = aes(label = occasion),
8                   show.legend = FALSE)
```



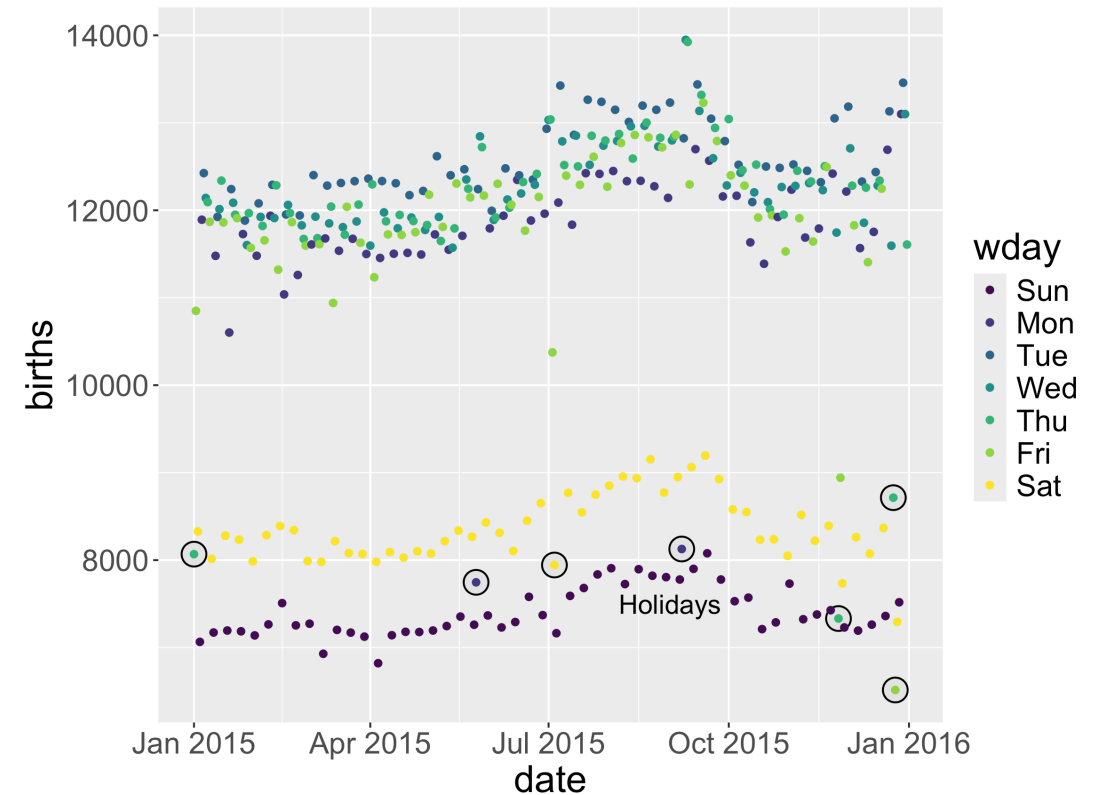
Context: Adding Holidays

```
1 ggplot(data = Births2015,  
2       mapping = aes(x = date, y = births,  
3                     color = wday)) +  
4   geom_point(size = 6, color = "black",  
5             data = holidays) +  
6   geom_point(size = 5, color = "grey90",  
7             data = holidays) +  
8   geom_point()
```



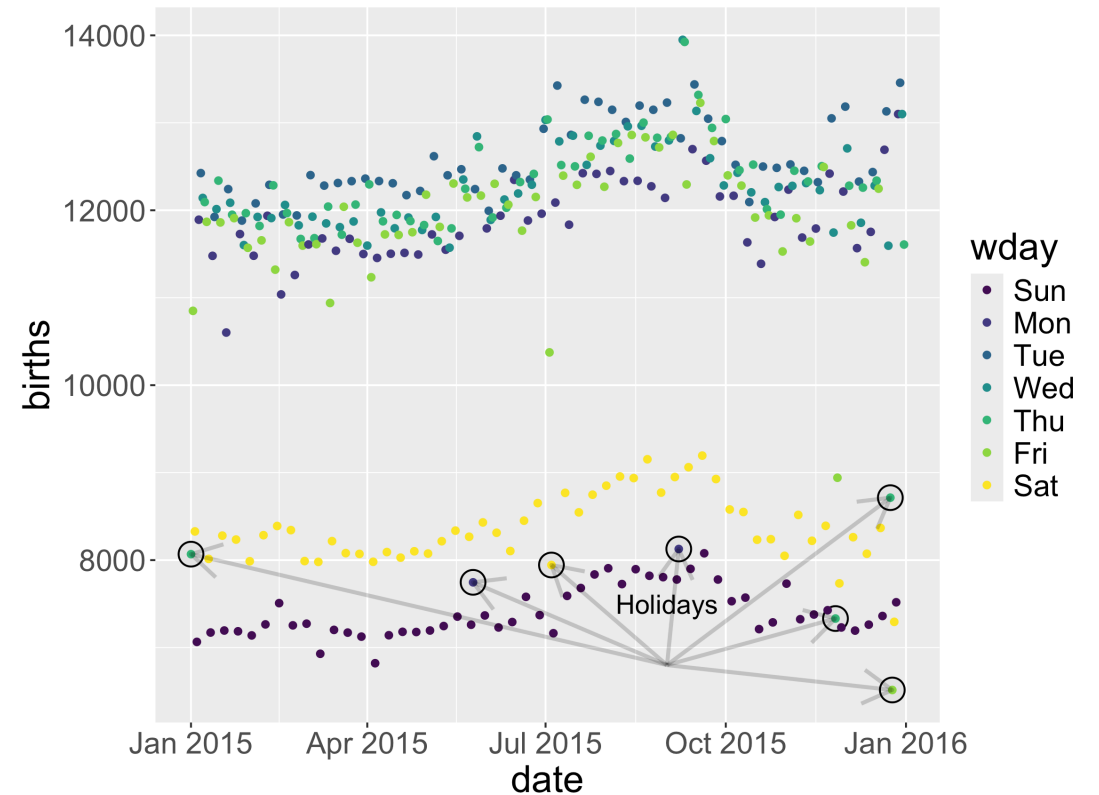
Context: Adding Holidays

```
1 ggplot(data = Births2015,  
2       mapping = aes(x = date, y = births,  
3                     color = wday)) +  
4   geom_point(size = 6, color = "black",  
5             data = holidays) +  
6   geom_point(size = 5, color = "grey90",  
7             data = holidays) +  
8   geom_point() +  
9   annotate("text", x = as_date("2015-09-01"),  
10          y = 7500, label = "Holidays",  
11          color="black", size=5)
```



Context: Adding Holidays

```
1 ggplot(data = Births2015,  
2       mapping = aes(x = date, y = births,  
3                     color = wday)) +  
4   geom_point(size = 6, color = "black",  
5             data = holidays) +  
6   geom_point(size = 5, color = "grey90",  
7             data = holidays) +  
8   geom_point() +  
9   annotate("segment", colour = "black",  
10          x = as_date("2015-09-01"),  
11          xend = holidays$date,  
12          y = 6800, yend = holidays$births,  
13          size = 1, alpha = 0.2, arrow = arrow()) +  
14   annotate("text", x = as_date("2015-09-01"),  
15          y = 7500, label = "Holidays",  
16          color = "black", size = 5)
```



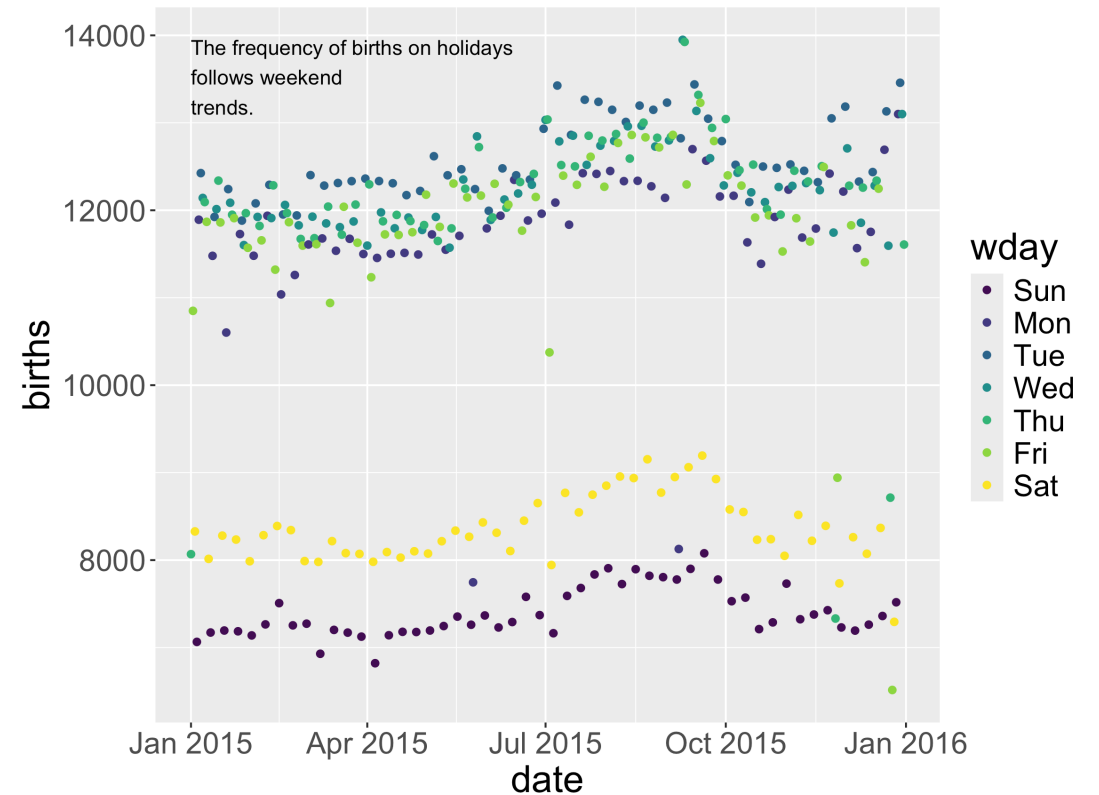
Context: Adding Holidays

```
1 # Create a story label
2 label_data <- data.frame(
3   date = ymd("2015-01-01"),
4   births = max(Births2015$births),
5   label = "The frequency of births on holidays \nfollows weekend \ntrends."
6 )
```

- What do you think “\n” does?

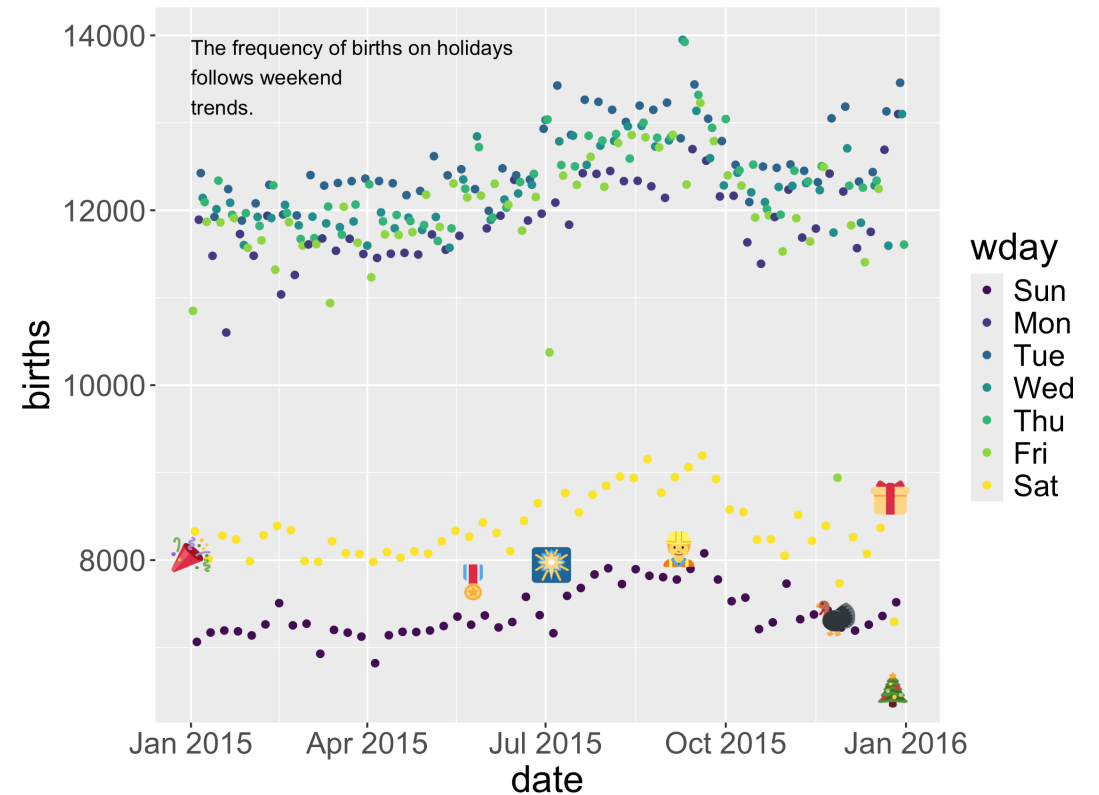
Context: Adding Holidays

```
1 ggplot(data = Births2015,  
2       mapping = aes(x = date, y = births,  
3                     color = wday)) +  
4   geom_point() +  
5   geom_text(mapping = aes(label = label),  
6             data = label_data,  
7             color = "black", vjust = "top",  
8             hjust = "left")
```



Context: Adding Holidays

```
1 # devtools::install_github("dill/emoGG")
2 library(emoGG)
3 ggplot(data = Births2015,
4       mapping = aes(x = date, y = births,
5                     color = wday)) +
6   geom_point() +
7   geom_text(mapping = aes(label = label),
8             data = label_data,
9             color = "black", vjust = "top",
10            hjust = "left") +
11   geom_emoji(data = holidays,
12             mapping = aes(emoji = emoji,
13                           x = date,
14                           y = births))
```

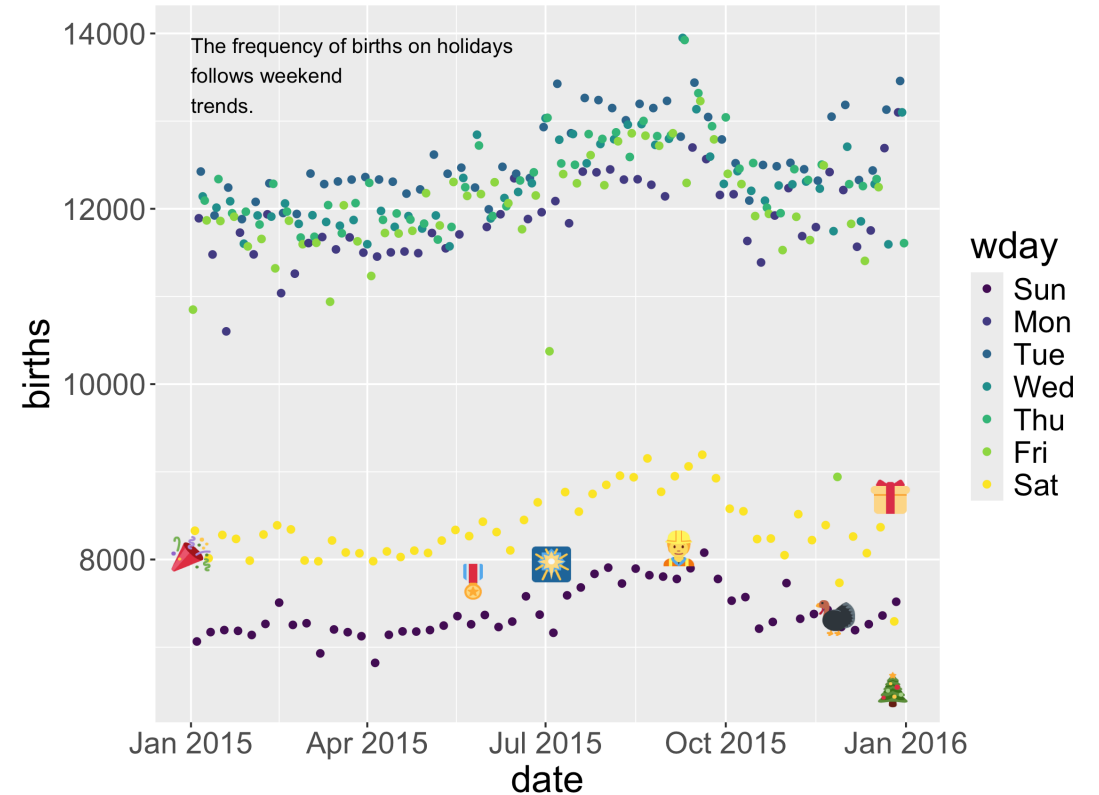


Context

And there are lots more ways to annotate your graph (shaded regions, spike lines...).

A few notes:

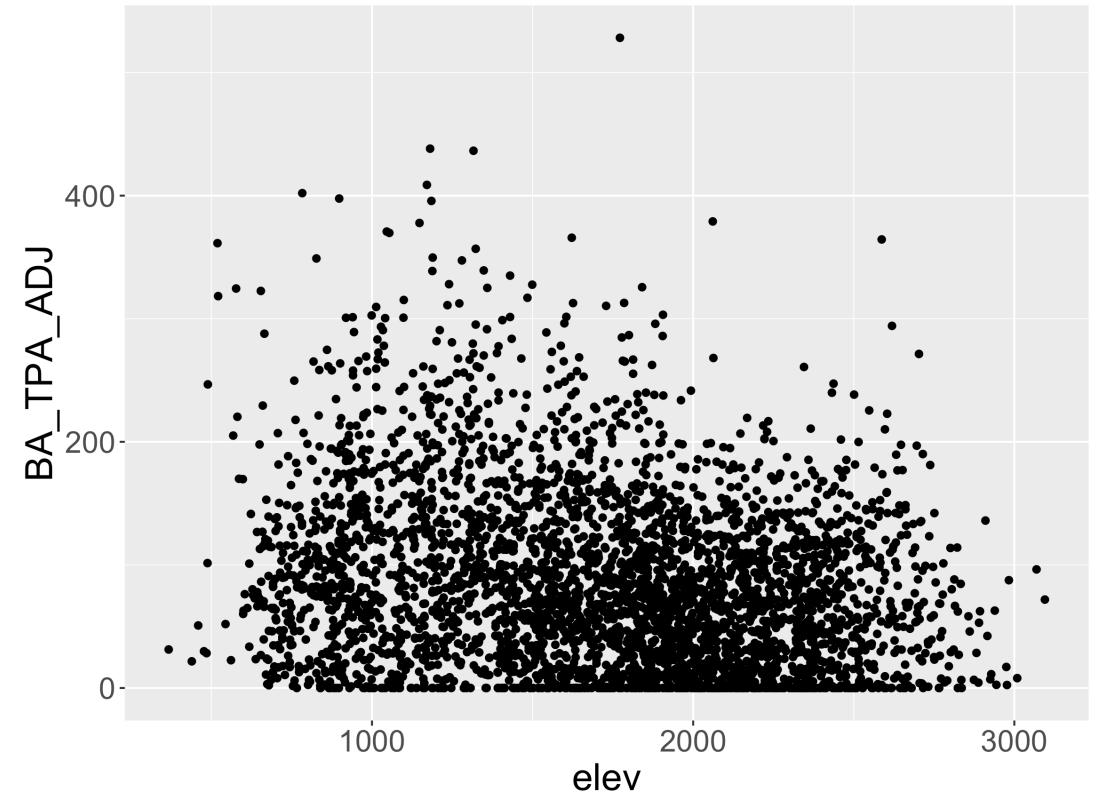
- Don't over do it!
- Like with selecting a **geom** or a **mapping** or a **scale**, try several out first.



Handling Over-plotting

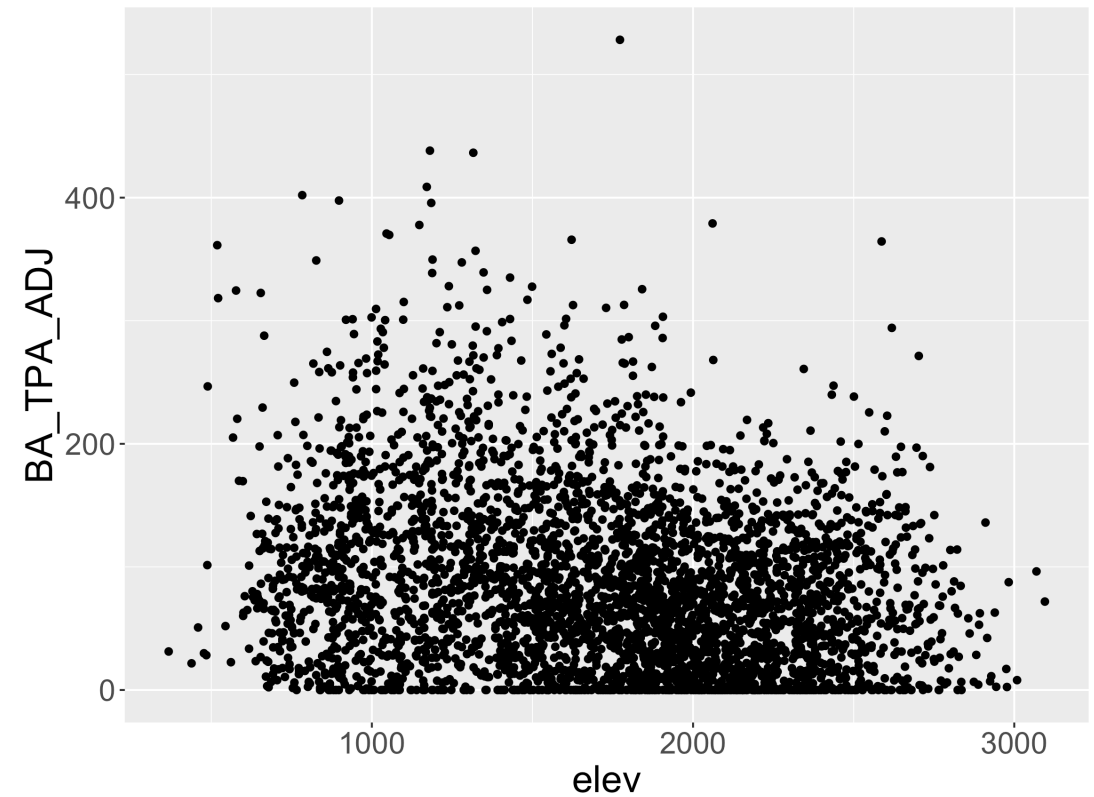
Let's return to the US Forest Inventory and Analysis Program Idaho data

```
1 dat <- readRDS("data/IDdata.rds")
2 idaho_plots <- dat$pltassgn
3 ggplot(data = idaho_plots,
4       mapping = aes(x = elev,
5                     y = BA_TPA_ADJ)) +
6   geom_point()
```



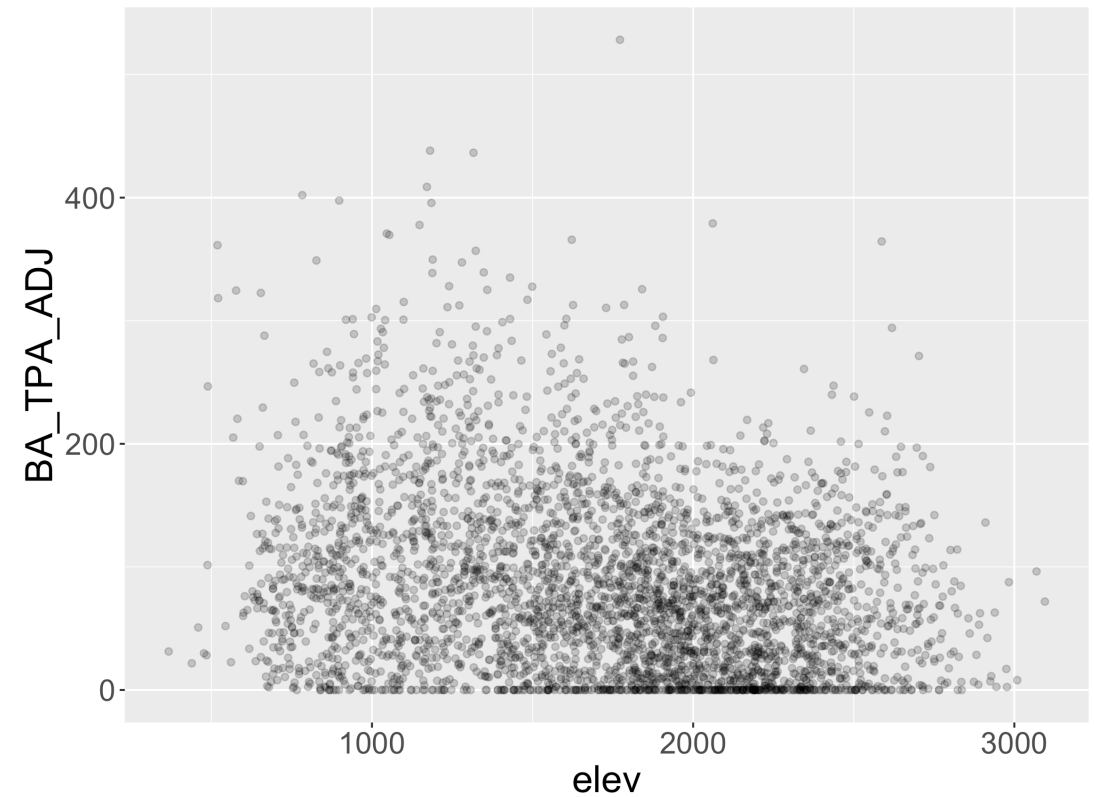
Handling Over-plotting

```
1 ggplot(data = idaho_plots,  
2       mapping = aes(x = elev,  
3                     y = BA_TPA_ADJ)) +  
4   geom_jitter()
```



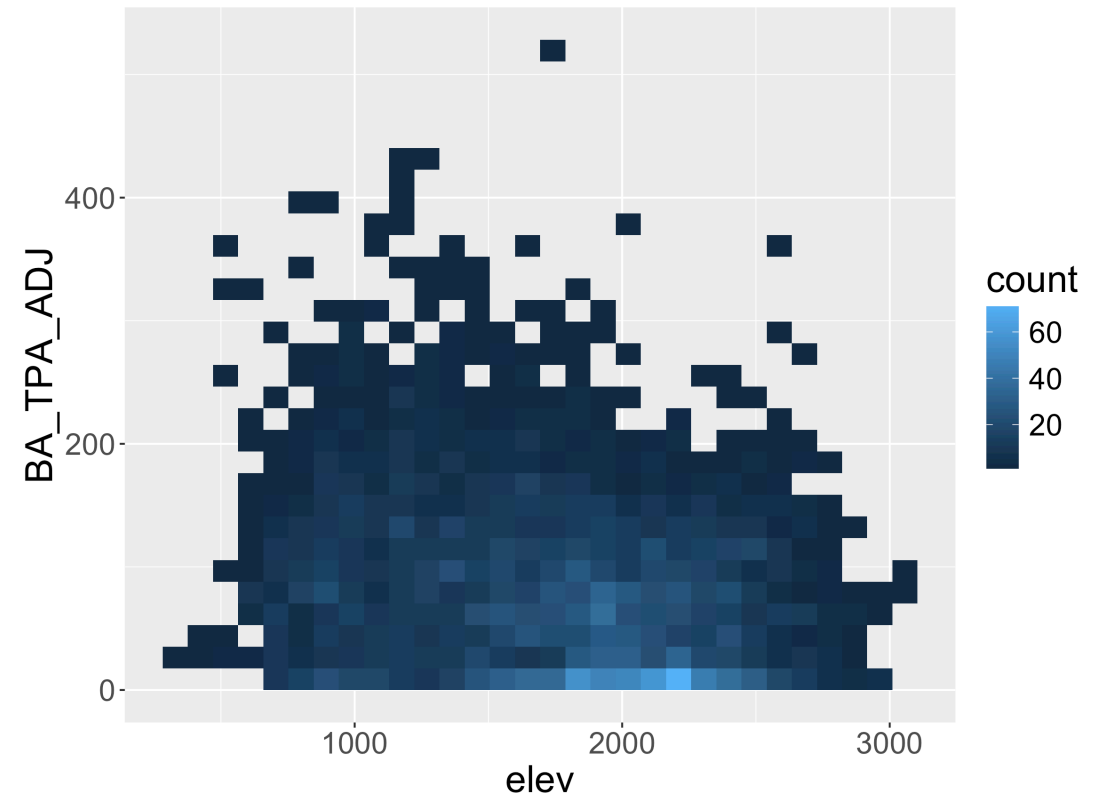
Handling Over-plotting

```
1 ggplot(data = idaho_plots,  
2       mapping = aes(x = elev,  
3                     y = BA_TPA_ADJ)) +  
4   geom_point(alpha = 0.2)
```



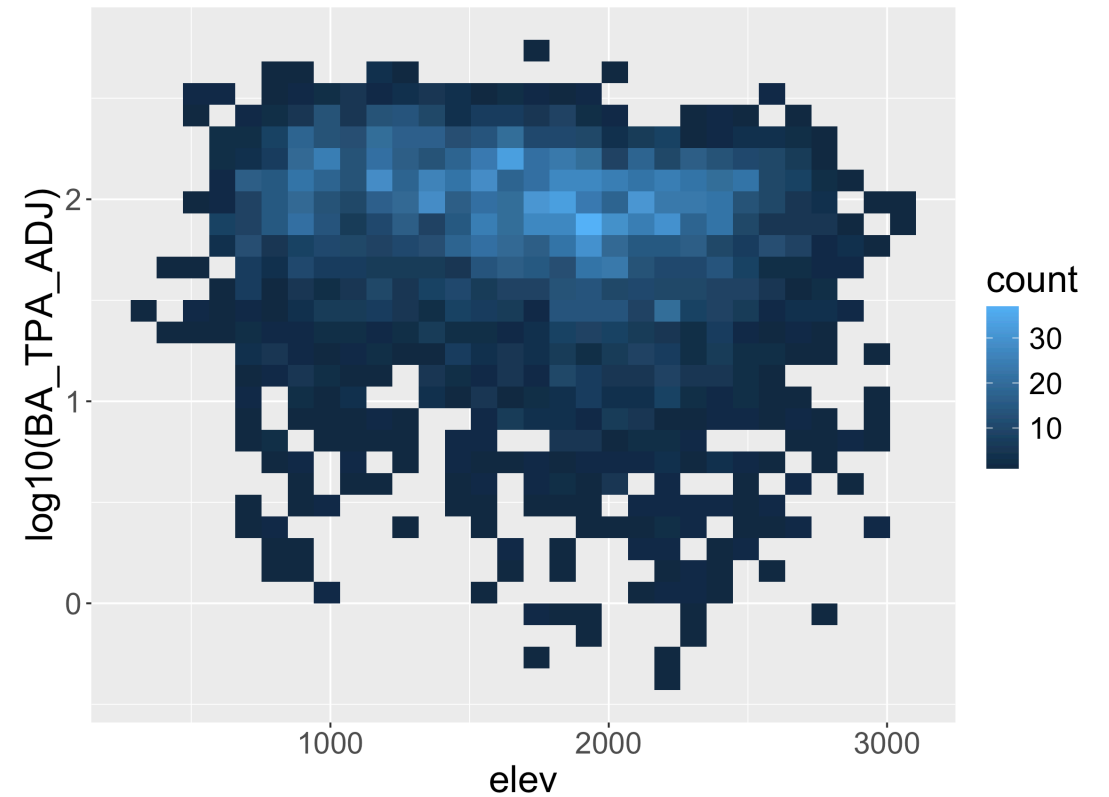
Handling Over-plotting

```
1 ggplot(data = idaho_plots,  
2       mapping = aes(x = elev,  
3                     y = BA_TPA_ADJ)) +  
4   geom_bin2d()
```



Handling Transformations

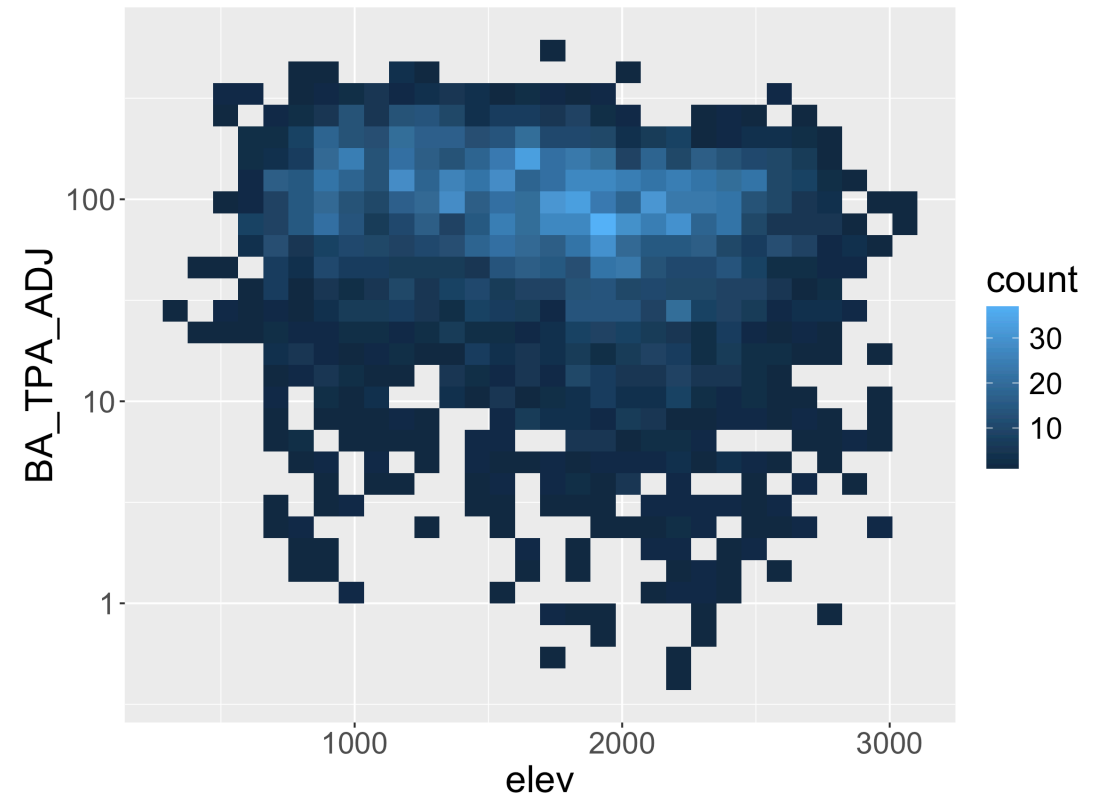
```
1 ggplot(data = idaho_plots,  
2       mapping = aes(x = elev,  
3                     y = log10(BA_TPA_ADJ))) +  
4   geom_bin2d()
```



- Transform variables directly.

Handling Transformations

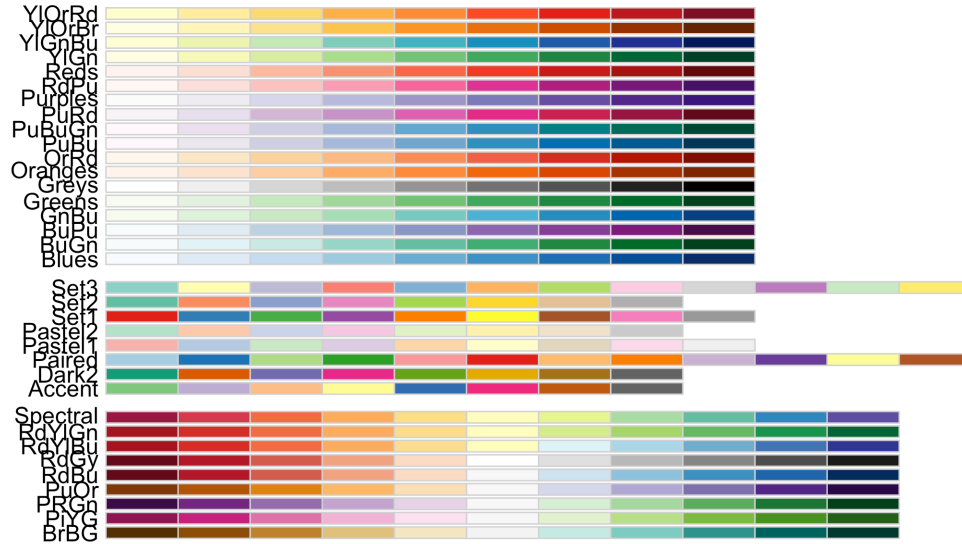
```
1 ggplot(data = idaho_plots,  
2       mapping = aes(x = elev,  
3                     y = BA_TPA_ADJ)) +  
4   geom_bin2d() +  
5   scale_y_log10()
```



- Transform scale.

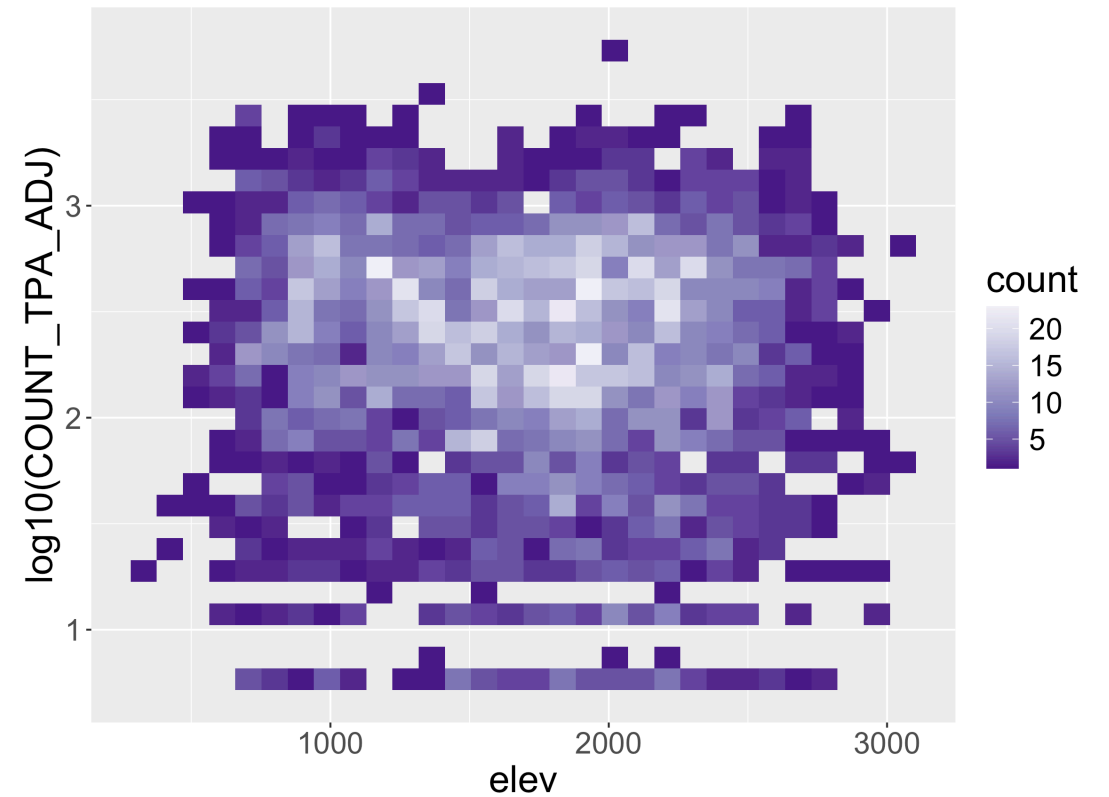
ColorBrewer

```
1 RColorBrewer::display.brewer.all()
```



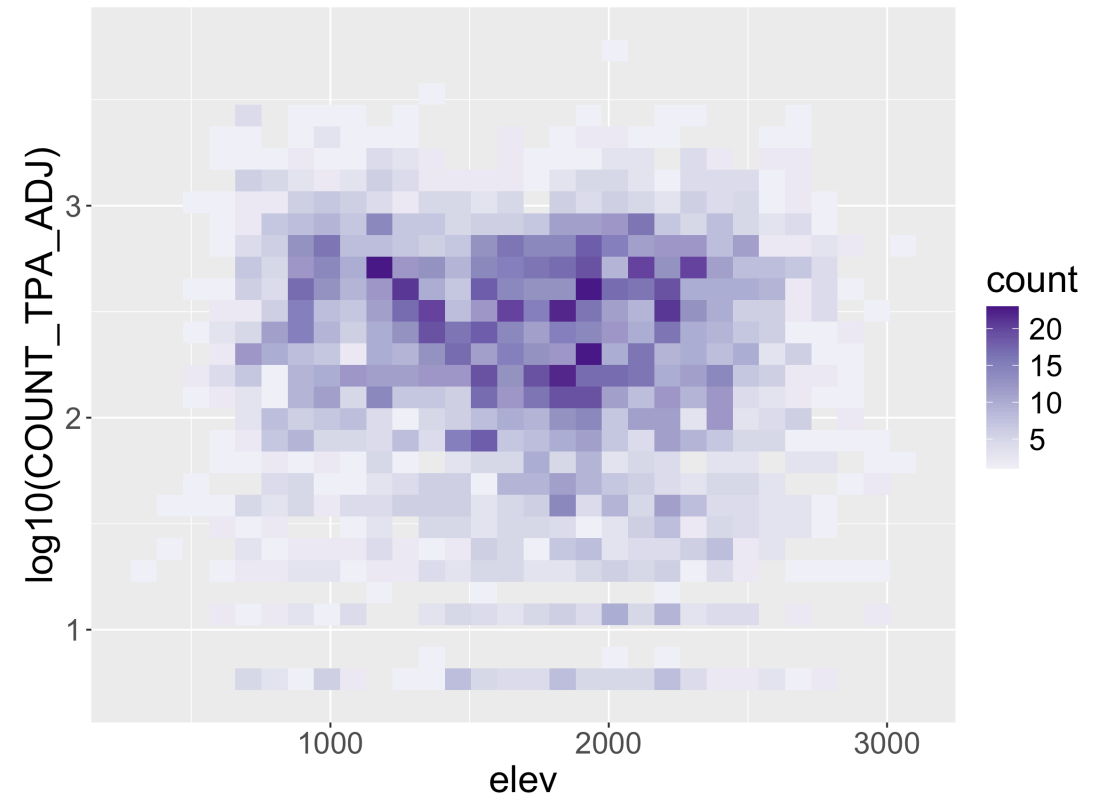
Color Options: Saturation

```
1 ggplot(data = idaho_plots,  
2       mapping = aes(x = elev,  
3                     y = log10(COUNT_TPA_ADJ))) +  
4   geom_bin2d() +  
5   scale_fill_distiller(palette = "Purples")
```



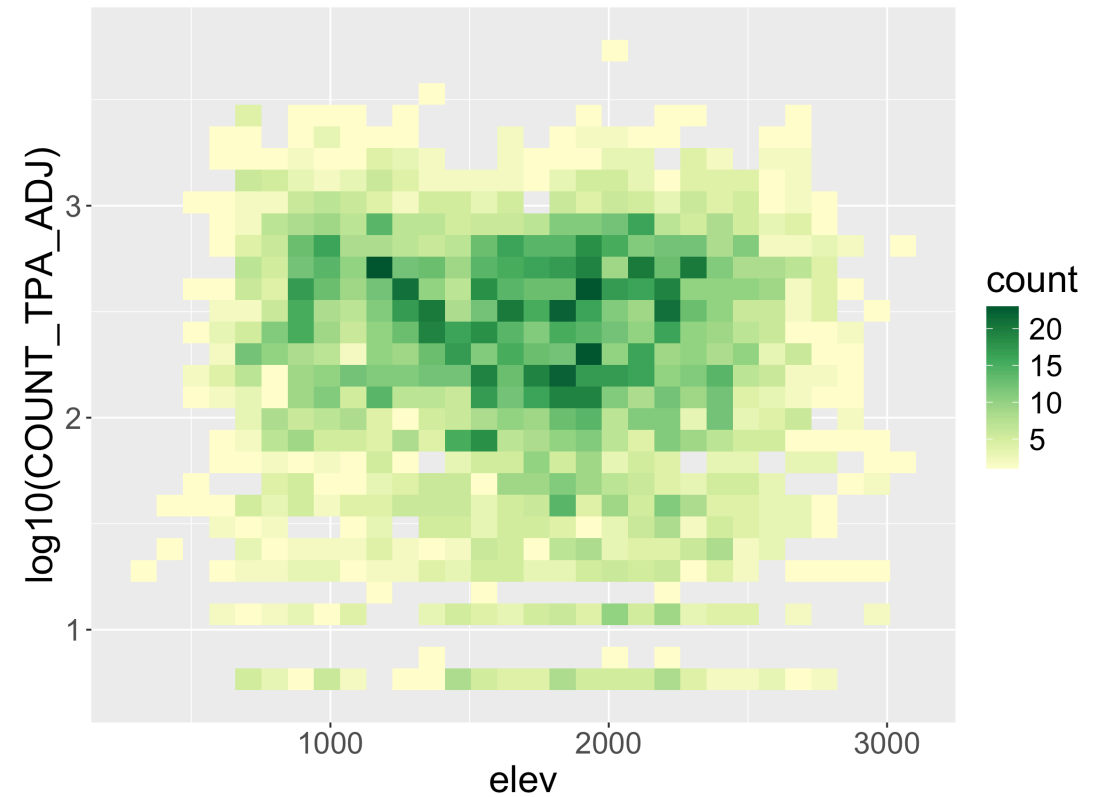
Color Options: Saturation

```
1 ggplot(data = idaho_plots,  
2       mapping = aes(x = elev,  
3                     y = log10(COUNT_TPA_ADJ))) +  
4   geom_bin2d() +  
5   scale_fill_distiller(palette = "Purples",  
6                       direction = 1)
```



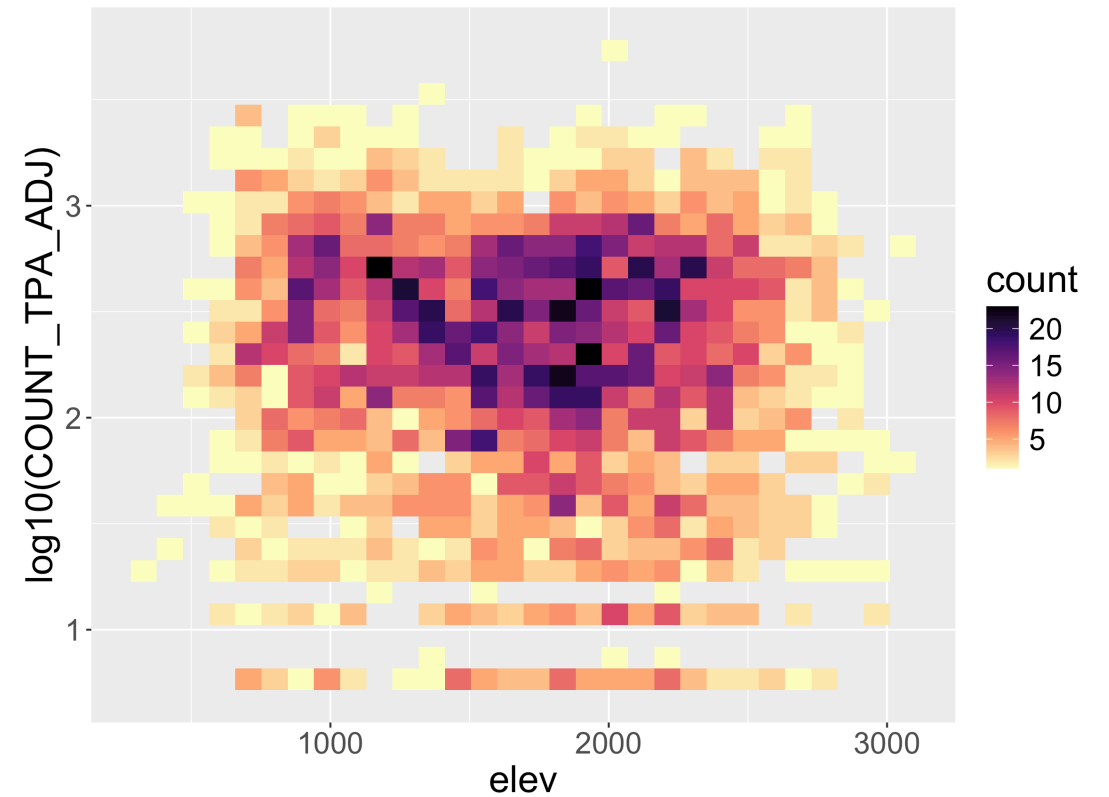
ColorBrewer YlGn palette

```
1 ggplot(data = idaho_plots,  
2       mapping = aes(x = elev,  
3                     y = log10(COUNT_TPA_ADJ))) +  
4   geom_bin2d() +  
5   scale_fill_distiller(palette = "YlGn",  
6                       direction = 1)
```



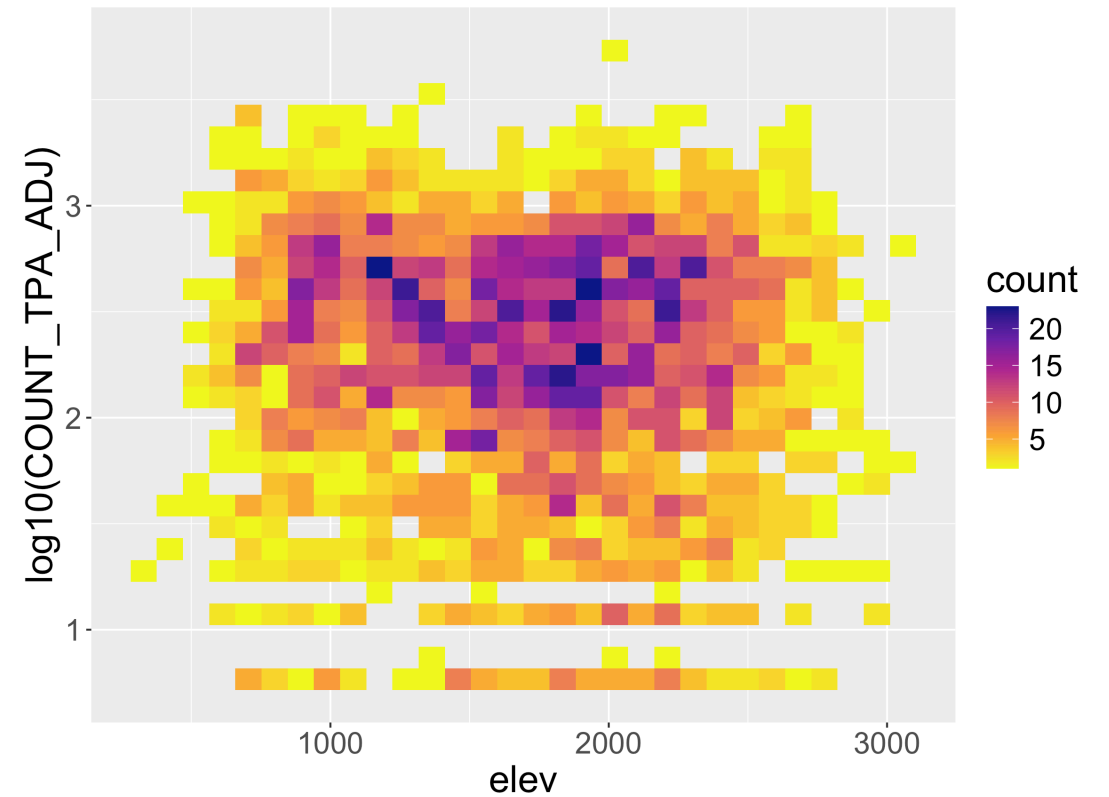
Viridis Palette

```
1 library(viridis)
2 ggplot(data = idaho_plots,
3       mapping = aes(x = elev,
4                     y = log10(COUNT_TPA_ADJ))) +
5   geom_bin2d() +
6   scale_fill_viridis_c(direction = -1,
7                         option = "A")
```



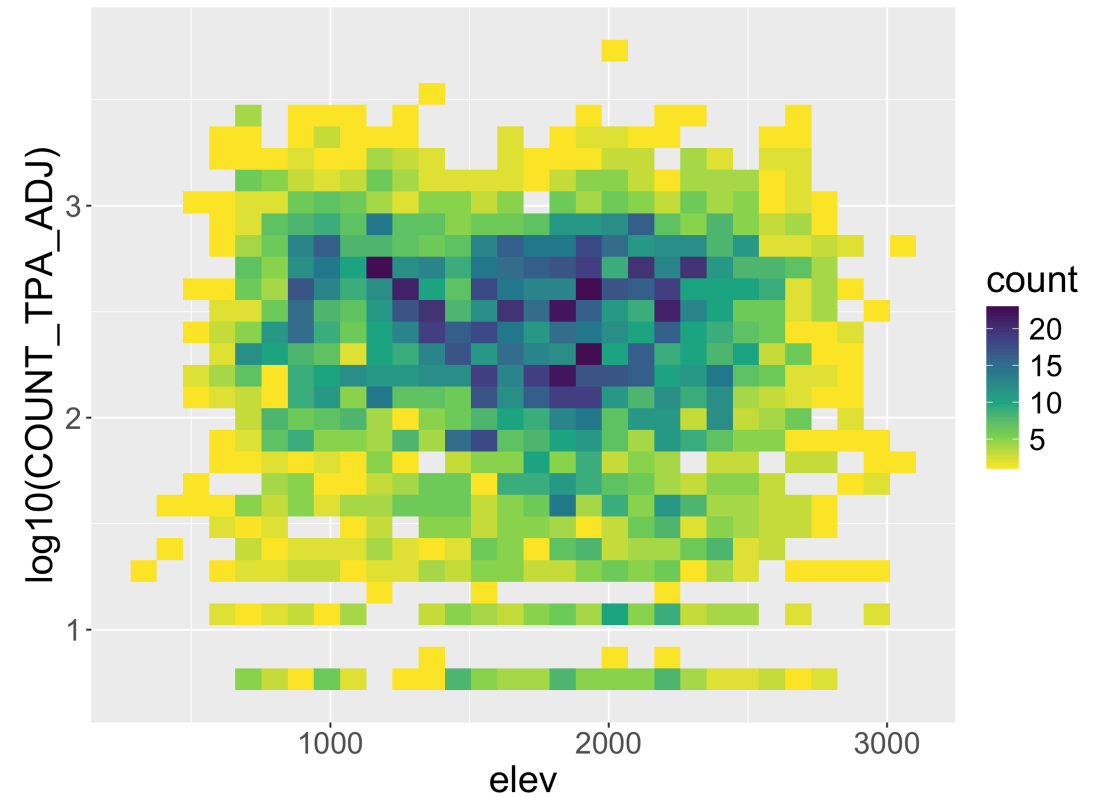
Viridis Palette

```
1 library(viridis)
2 ggplot(data = idaho_plots,
3       mapping = aes(x = elev,
4                     y = log10(COUNT_TPA_ADJ))) +
5   geom_bin2d() +
6   scale_fill_viridis_c(direction = -1,
7                         option = "C")
```



Viridis Palette

```
1 library(viridis)
2 ggplot(data = idaho_plots,
3       mapping = aes(x = elev,
4                     y = log10(COUNT_TPA_ADJ))) +
5   geom_bin2d() +
6   scale_fill_viridis_c(direction = -1,
7                         option = "D")
```



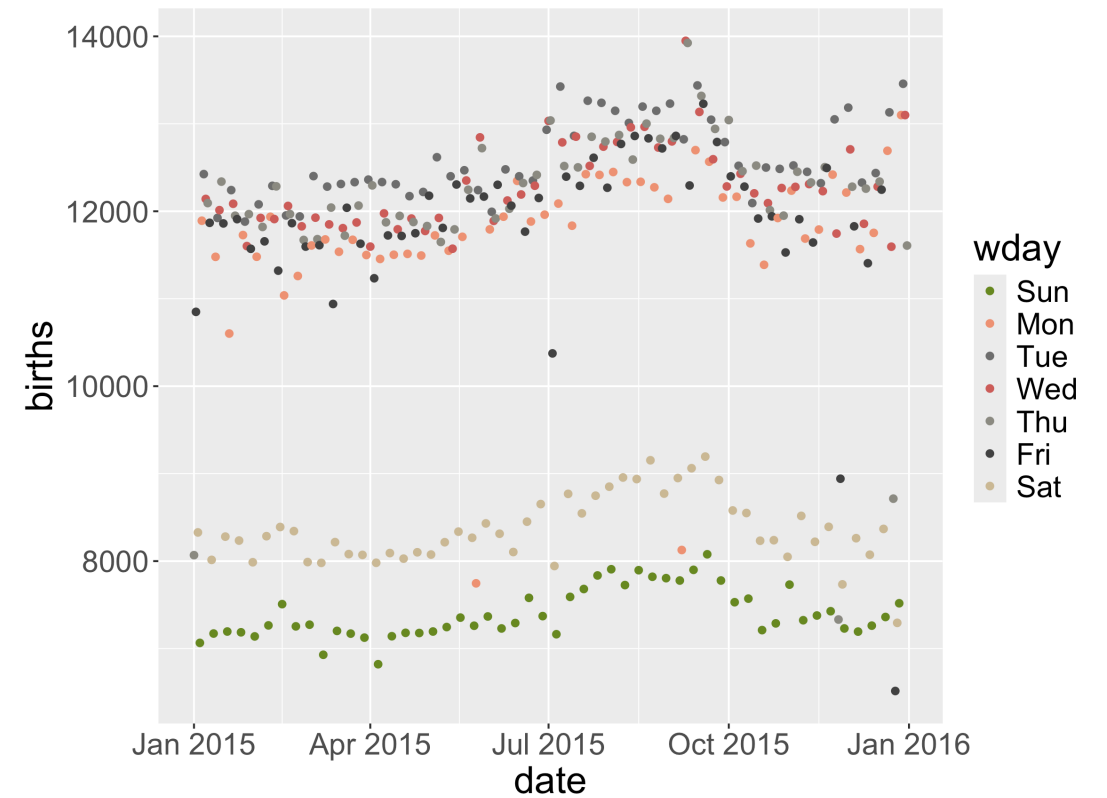
Color Options: Hue

```
1 colors()
```

```
[1] "white"           "aliceblue"       "antiquewhite"
[4] "antiquewhite1"   "antiquewhite2"   "antiquewhite3"
[7] "antiquewhite4"   "aquamarine"      "aquamarine1"
[10] "aquamarine2"     "aquamarine3"     "aquamarine4"
[13] "azure"           "azure1"          "azure2"
[16] "azure3"          "azure4"          "beige"
[19] "bisque"          "bisque1"         "bisque2"
[22] "bisque3"         "bisque4"         "black"
[25] "blanchedalmond"  "blue"            "blue1"
[28] "blue2"           "blue3"           "blue4"
[31] "blueviolet"      "brown"           "brown1"
[34] "brown2"          "brown3"          "brown4"
[37] "burlywood"       "burlywood1"      "burlywood2"
[40] "burlywood3"      "burlywood4"      "cadetblue"
- -
```

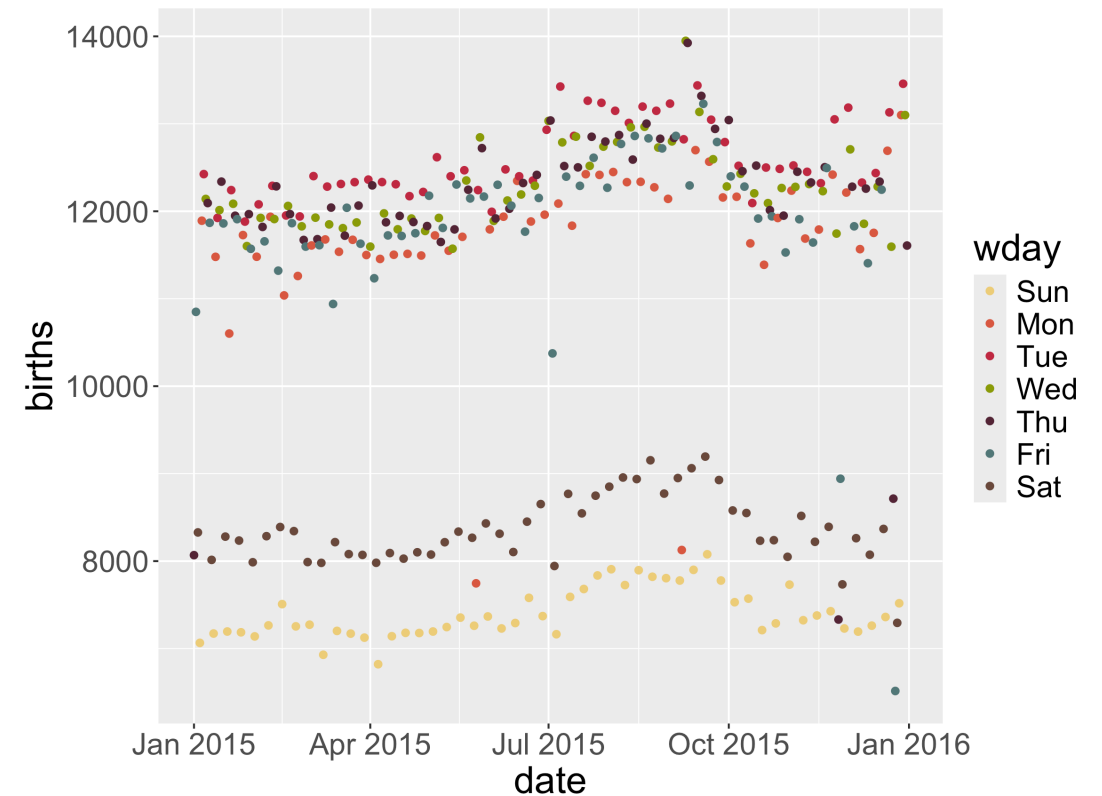
Color Options: Hue

```
1 ggplot(data = Births2015,  
2       mapping = aes(x = date, y = births,  
3                     color = wday)) +  
4   geom_point() +  
5   scale_color_manual(values =  
6                     sample(colors(), 7))
```



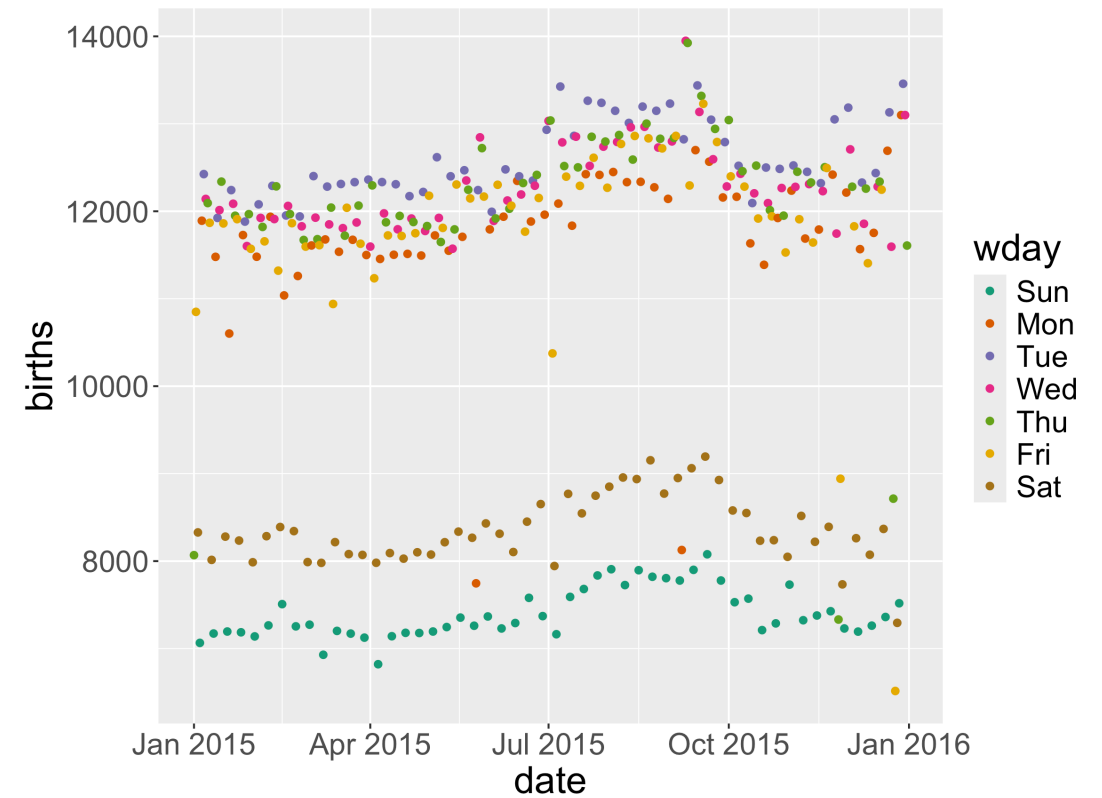
Color Options: Hue

```
1 ggplot(data = Births2015,  
2       mapping = aes(x = date, y = births,  
3                     color = wday)) +  
4   geom_point() +  
5   scale_color_manual(values=  
6     c("#ECD078", "#D95B43",  
7       "#C02942", "#8A9B0F",  
8       "#542437", "#53777A",  
9       "#6A4A3C"))
```



ColorBrewer Hue

```
1 ggplot(data = Births2015,  
2       mapping = aes(x = date, y = births,  
3                     color = wday)) +  
4   geom_point() +  
5   scale_color_brewer(palette = "Dark2")
```



Themes

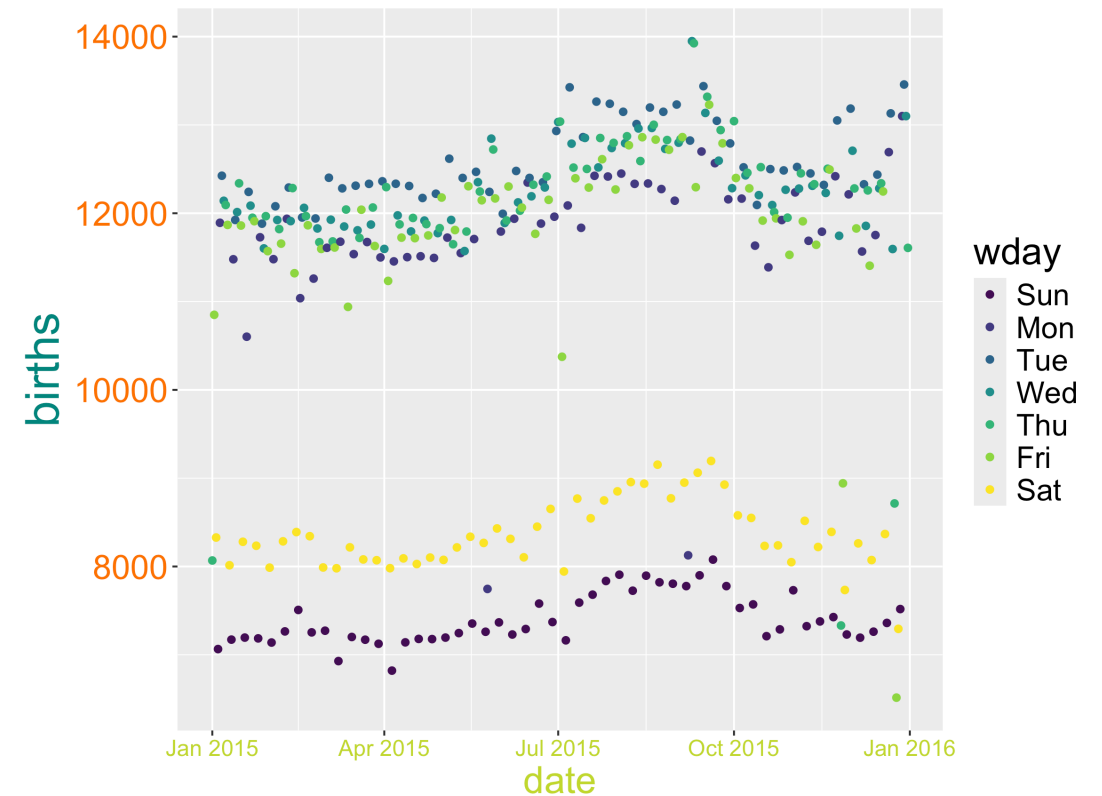
```
1 # ?theme
```

- Can override specific aspects of the theme
 - EX: `+ theme(legend.position = "bottom")`
- Can globally set theme options:

```
1 theme_update(text = element_text(size = 20))
```

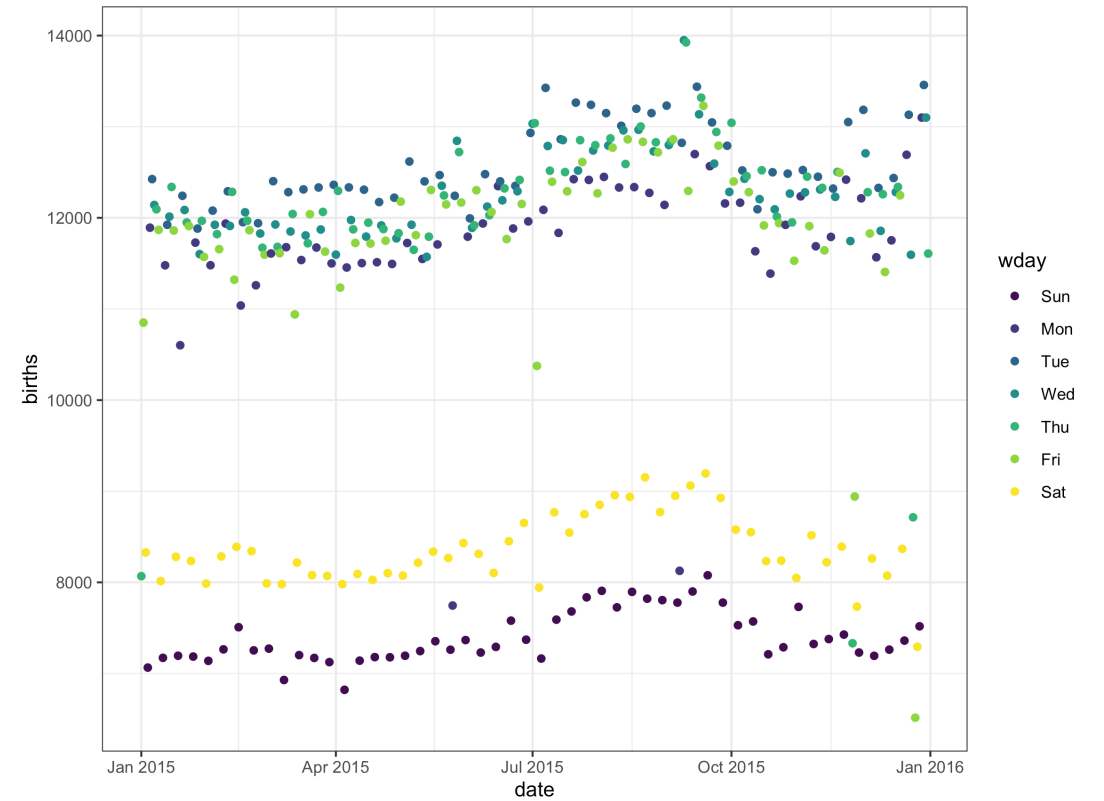
Themes

```
1 ggplot(data = Births2015,  
2       mapping = aes(x = date, y = births,  
3                     color = wday)) +  
4   geom_point() +  
5   theme(axis.title.x =  
6         element_text(color = "#C1D82F", size = 20),  
7         axis.title.y =  
8         element_text(color = "#00857D", size = 25),  
9         axis.text.x =  
10        element_text(color = "#C1D82F", size = 12),  
11        axis.text.y =  
12        element_text(color = "#FF7401", size = 18))
```



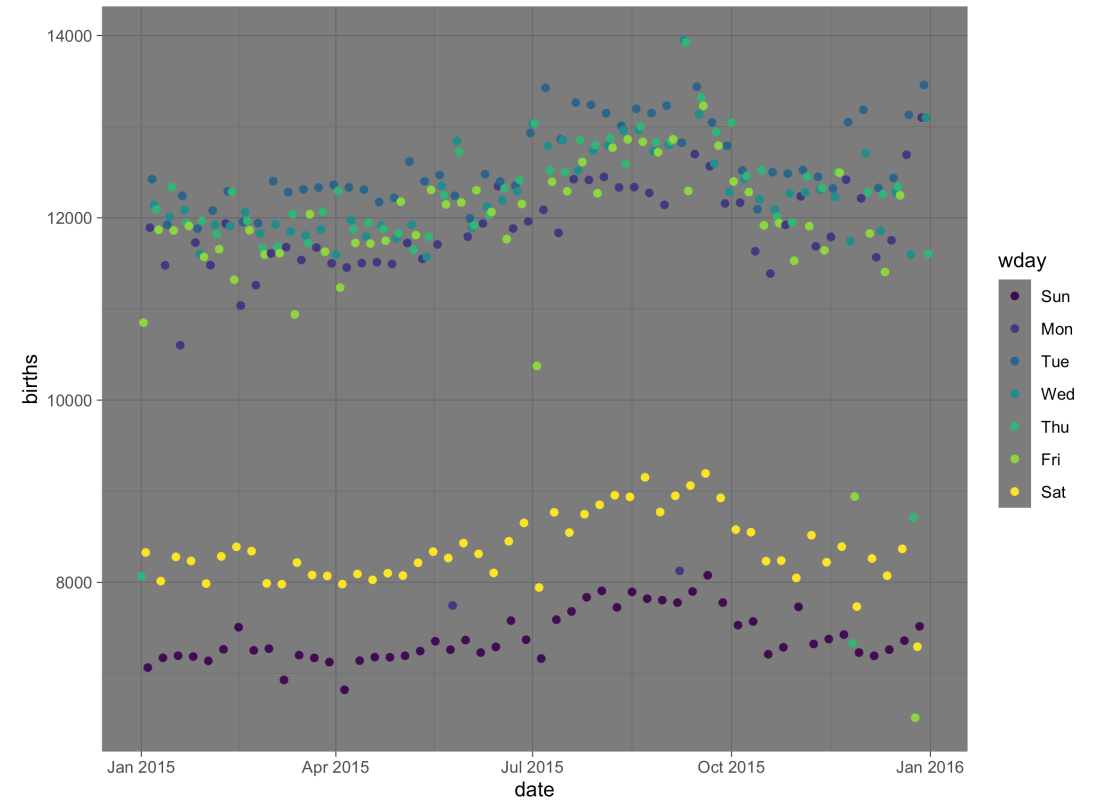
Built-in Themes

```
1 ggplot(data = Births2015,  
2       mapping = aes(x = date, y = births,  
3                     color = wday)) +  
4   geom_point() +  
5   theme_bw()
```



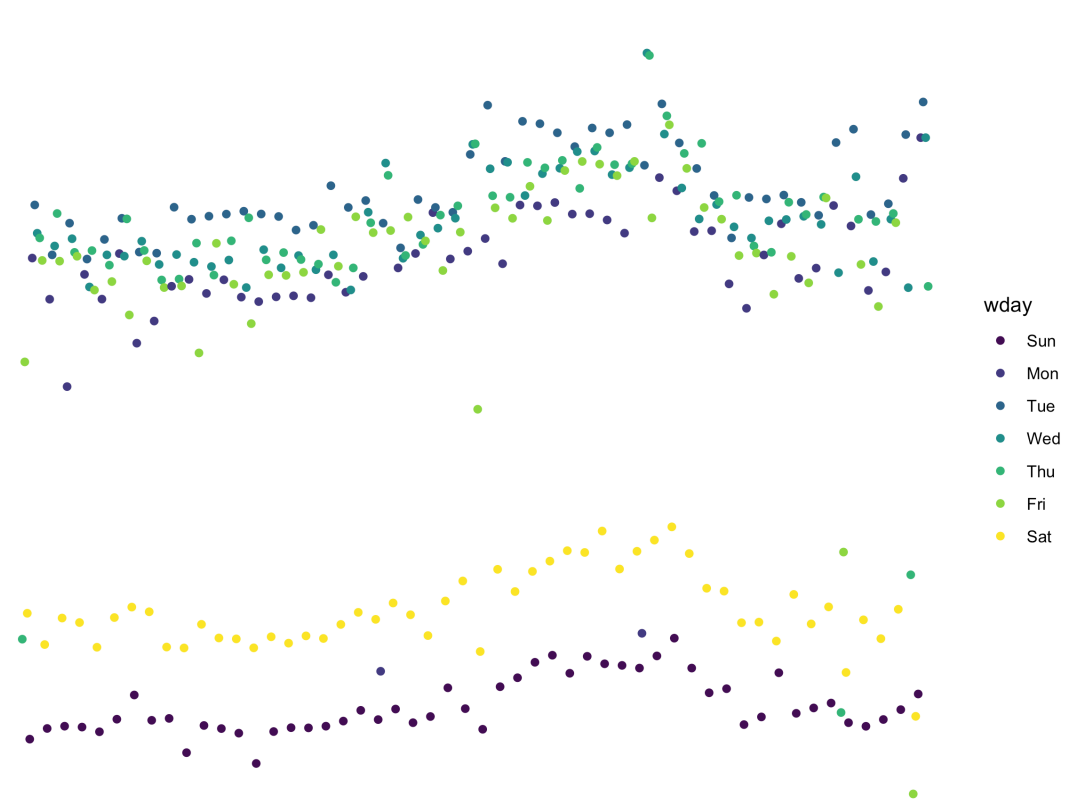
Built-in Themes

```
1 ggplot(data = Births2015,  
2       mapping = aes(x = date, y = births,  
3                     color = wday)) +  
4   geom_point() +  
5   theme_dark()
```



Built-in Themes

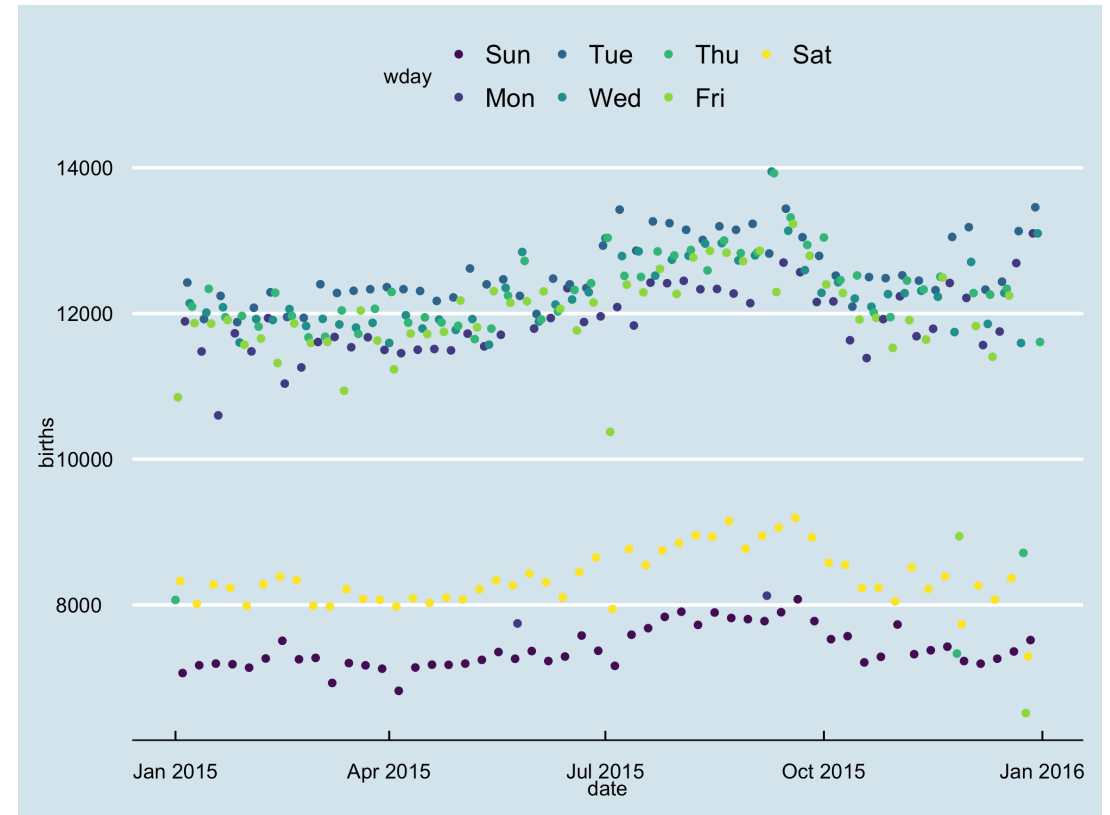
```
1 ggplot(data = Births2015,  
2       mapping = aes(x = date, y = births,  
3                     color = wday)) +  
4   geom_point() +  
5   theme_void()
```



- Useful for maps and pie charts!

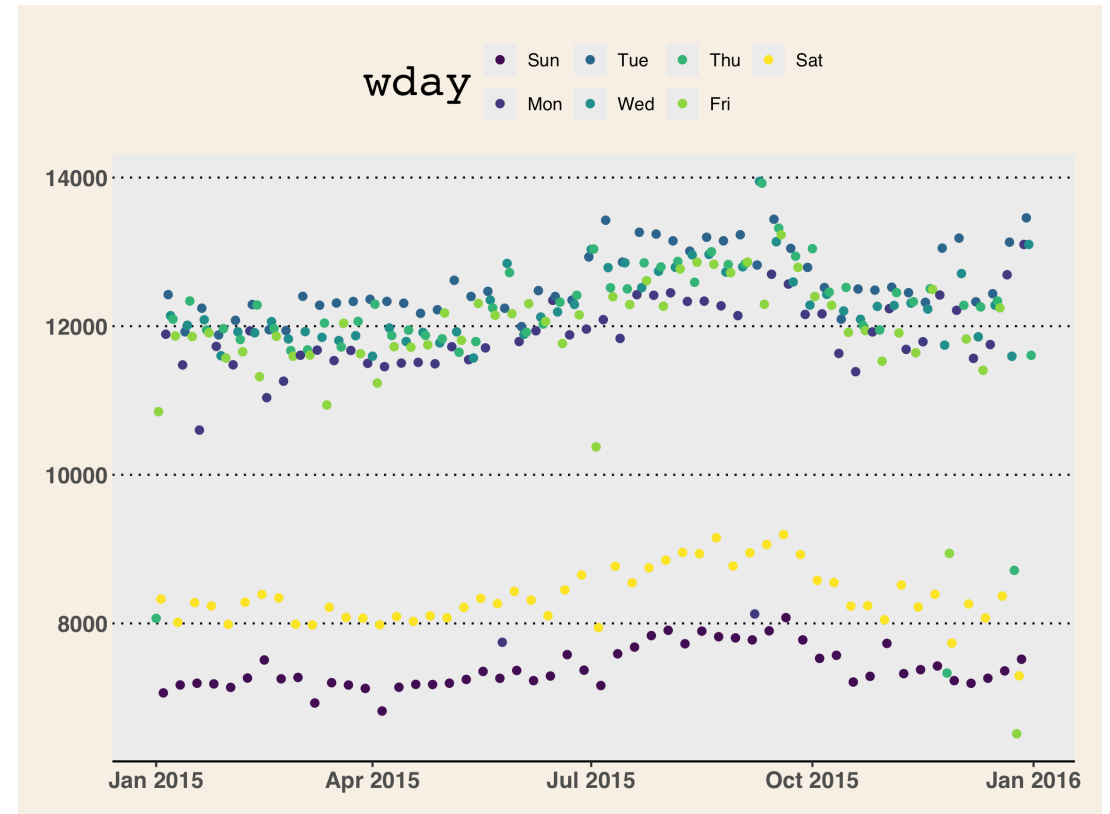
Additional Themes Package: ggthemes

```
1 library(ggthemes)
2 ggplot(data = Births2015,
3       mapping = aes(x = date, y = births,
4                     color = wday)) +
5   geom_point() +
6   theme_economist()
```



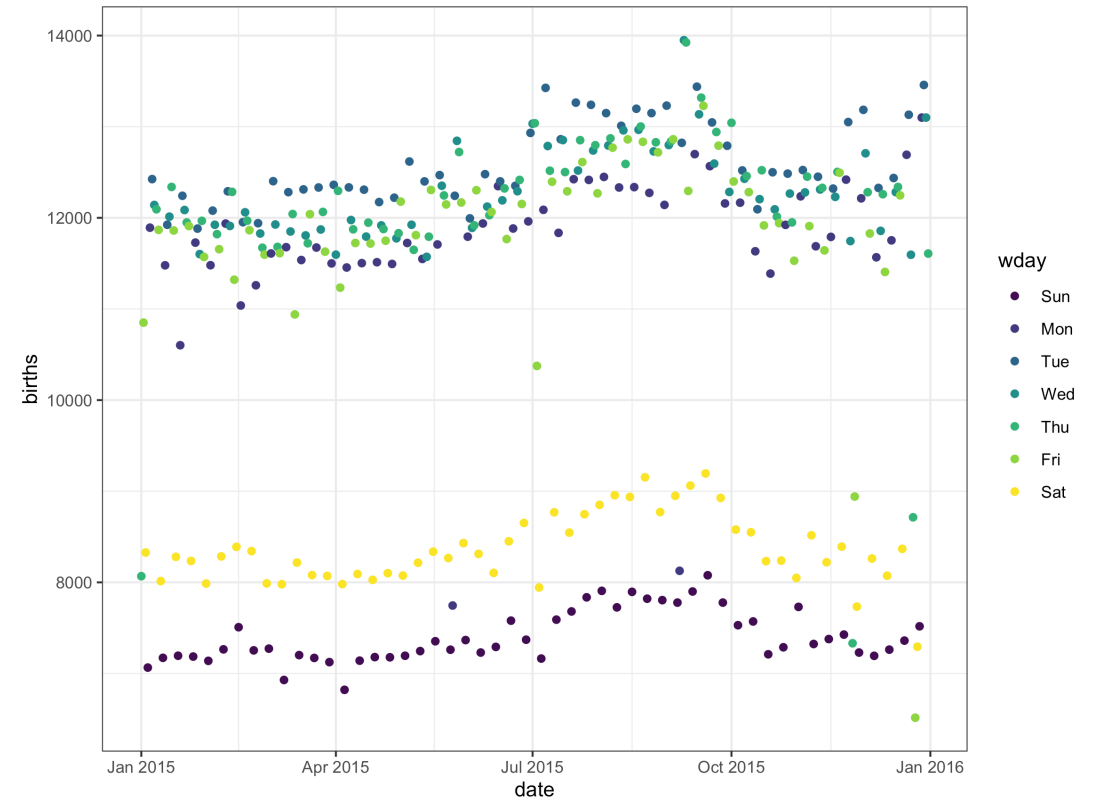
Additional Themes Package

```
1 ggplot(data = Births2015,  
2       mapping = aes(x = date, y = births,  
3                     color = wday)) +  
4   geom_point() +  
5   theme_wsj()
```



Saving/exporting your figures

```
1 p1 <- ggplot(data = Births2015,  
2             mapping = aes(x = date, y = births,  
3                           color = wday)) +  
4   geom_point() +  
5   theme_bw()  
6  
7 p1
```



```
1 ggsave(filename = "my_nice_figure.png",  
2         plot = p1,  
3         width = 10,  
4         height = 6,  
5         units = "in")
```

Reproducible Workflow

- One where if you shared your data and work with someone else, they could reproduce your results.
- Not the same as **replication**: Where someone collects new data following your same design to see if they get the same results.
- **Quarto** documents allow us to include our **R** code, output, and narrative in the same place.
 - Load the **raw** data.
 - Be transparent about all the analysis steps.
 - Even if you don't showcase the **R** code in the output file, it is contained in the **qmd** file.



Creating reproducible examples with reprex



Why do I need to learn to create reproducible technical examples?

- So that you can ask and answer questions in our class Slack Workspace or Stack Overflow or other R help sites!

First, let's take a look at some bad examples!

What is wrong with this coding question?

I am trying to create a plot and I can't get the bars to do what I want them to. Help?!

What is wrong with this coding question?

I want to do the following but it isn't working:

```
thing <- read.csv("long/file/path/thing.csv")
```

```
ggplot(thing, aes(x = factor(that))) + geom_bar()
```

Help?!

What is wrong with this coding question?

I want to reorder the bars of my plot but can't get it working. Help!

```
1 library(tidyverse)
2 library(palmerpenguins)
3
4 penguins <- penguins %>%
5   group_by(species) %>%
6   mutate(mean_flipper = mean(flipper_length_mm)) %>%
7   ungroup() %>%
8   mutate(long = case_when(flipper_length_mm < mean(flipper_length_mm) ~ "no",
9                           flipper_length_mm >= mean(flipper_length_mm) ~ "yes"))
10
11 penguins %>%
12   ggplot(mapping = aes(x = factor(species))) +
13   geom_bar()

1 penguins %>%
2   count(species)
```

What is wrong with this coding question?

I want to reorder the bars of my plot but can't get it working. Help!

```
1 rm(list = ls())
2
3 library(tidyverse)
4 library(palmerpenguins)
5
6 penguins %>%
7   ggplot(mapping = aes(x = factor(species))) +
8   geom_bar()
```

What makes a good coding question?

- It uses a **minimal** dataset to reproduce the issue.
- It includes the **shortest** amount of **runnable** code necessary to reproduce the issue.
- It doesn't wreak havoc on other people's computers.
- It includes code **and output** so that others don't have to run it!
- It includes any necessary information on the used packages, R version, system, etc.
 - Should not be a concern for our class Slack since we are all on the same RStudio Server.
 - Can use `packageVersion("tidyverse")` or `sessionInfo()` to find this information.

Minimal Dataset: two good options

Create a toy data frame.

```
1 dat <- data.frame(animal = c("cat", "dog", "mouse"),  
2                   weight = c(5, 10, 0.5))  
3 dat
```

```
  animal weight  
1   cat    5.0  
2   dog   10.0  
3 mouse    0.5
```

Use a built-in dataset or a dataset from a particular package.

```
1 library(palmerpenguins)  
2 penguins
```

A tibble: 344 × 10

	species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g
	<fct>	<fct>	<dbl>	<dbl>	<int>	<int>
1	Adelie	Torgersen	39.1	18.7	181	3750
2	Adelie	Torgersen	39.5	17.4	186	3800
3	Adelie	Torgersen	40.3	18	195	3250
4	Adelie	Torgersen	NA	NA	NA	NA
5	Adelie	Torgersen	36.7	19.3	193	3450
6	Adelie	Torgersen	39.3	20.6	190	3650
7	Adelie	Torgersen	38.9	17.8	181	3625
8	Adelie	Torgersen	39.2	19.6	195	4675
9	Adelie	Torgersen	34.1	18.1	193	3475
10	Adelie	Torgersen	42	20.2	190	4250

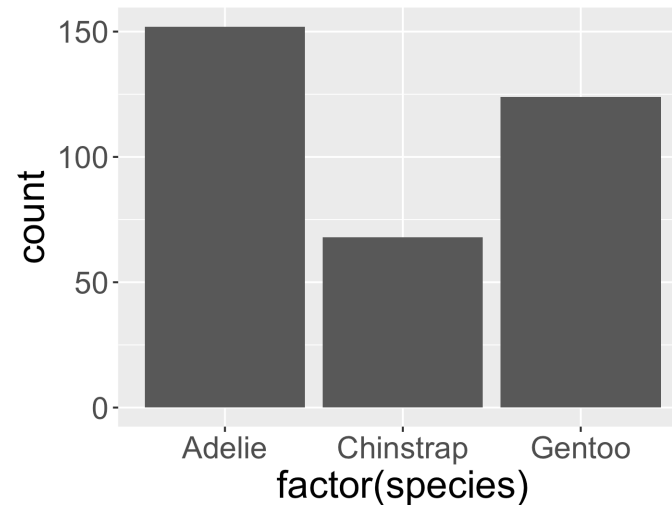
 334 more rows

Minimal Code

Include the **necessary** libraries.

Test run the code in a restarted R session to make sure it is runnable!

```
1 library(tidyverse)
2 library(palmerpenguins)
3
4 penguins %>%
5   ggplot(mapping = aes(x = factor(species))) +
6   geom_bar()
```



Make sure your code is copy-and-paste-able!

Don't copy from the console.

```
1 > library(tidyverse)
2 > library(palmerpenguins)
3 >
4 > penguins %>%
5 +   ggplot(mapping = aes(x = factor(species))) +
6 +   geom_bar()
```

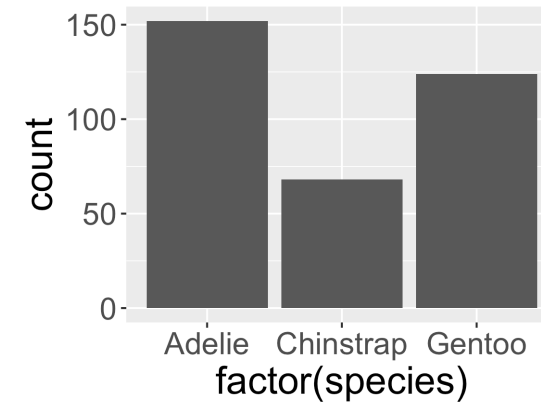
Make sure your code is copy-and-paste-able!

```
> library(tidyverse)
> library(palmerpenguins)
>
> penguins %>%
+   ggplot(mapping = aes(x = factor(species)))
+
+   geom_bar()
```

Now we have our reproducible example:

How can I reorder the bars in the ggplot to go from the most frequent to the least frequent category?

```
1 library(tidyverse)
2 library(palmerpenguins)
3
4 penguins %>%
5   ggplot(mapping = aes(x = factor(species))) +
6   geom_bar()
```



How can we easily share it?

- Using the `reprex()` function in the `reprex` package.

reprex Practice Time!

But first: Q: What is an R script file?

- A text file for entering R commands.

Q: How is an R script file different from a Quarto or RMarkdown document?

- You only put code in an R script.
- If you add any text you must comment it out with #.
- Think of it as a single R chunk that you won't knit into an output document.
- Useful when writing a lot of code and want to compartmentalize.

reprex Practice Time!

1. In **Session**, select “Clear Workspace” and then “Restart R”.
2. Open a script file and include in the top line:

```
1 library(reprex)
```

3. Put the code you want to use in the script file and make sure it runs.

```
1 library(tidyverse)
2 library(palmerpenguins)
3
4 penguins %>%
5   ggplot(mapping = aes(x = factor(species))) +
6   geom_bar()
```

4. Surround the code with `reprex({ ... }, venue = "slack")` and run it.
5. An md file will pop up. Copy all the contents of that file.
6. Head over to the **#coding-qa** channel and paste in the contents as a reply to the **reprex** practice message. A text box will pop-up and select “Apply”.
7. Above your code, type your question. Then hit “Send”.



Next week

- Learn to make animated figures!
- Hone our data wrangling skills
- P-set 1 due on Thursday 2/12